

Space Computing Constellation: System Architecture, Implementations, and Challenges

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Abstract—Low-Earth-orbit (LEO) satellite constellations have experienced rapid growth in recent years, driven by their potential to deliver global, high-bandwidth Internet services with low latency. Beyond connectivity, LEO constellations also offer promising opportunities to enable in-orbit processing of space-native data to support a wide range of emerging space applications. In this context, the concept of space computing has been proposed, a paradigm that seamlessly integrates networking and computing to provide computing-as-a-service anytime and anywhere in space. However, the inherent characteristics of satellite constellations, such as dynamic network topologies, constrained system resources, and the harsh space environment, pose significant challenges in achieving this vision. This article outlines the system architecture and the key enabling technologies for space computing, including spaceborne computers, laser communications, spaceborne routers, distributed operating systems, and onboard artificial intelligence (AI). We also present the implementation of an open space computing platform, the 3-body computing constellation, along with the in-orbit experimental results that demonstrate the advantages of multisatellite distributed computing. Furthermore, we outline future research directions essential for advancing toward a truly interconnected, autonomous, and intelligent space computing system.

Index Terms—Distributed operating system, laser communications, onboard artificial intelligence (AI), open platform, space computing, spaceborne computer, spaceborne router.

I. INTRODUCTION

IN RECENT years, satellite technology has advanced rapidly, driven by the reduction in the manufacturing cost of satellite components and the growing availability of ride-sharing launch opportunities. These developments have

significantly reduced the cost of satellite deployment, particularly in low Earth orbit (LEO) [1]. As a result, major organizations and commercial enterprises around the world have proposed or launched LEO mega-constellations, consisting of hundreds to thousands of satellites, to deliver low-latency, high-throughput broadband services on a global scale [2]. Prominent examples of such constellations include SpaceX's Starlink, London-based OneWeb, Amazon's Kuiper, Canada's Telesat, and China's Qianfan.

Another promising opportunity enabled by mega-constellations is their potential to provide computing services anytime and anywhere, provided that the satellites are equipped with more powerful computing resources [3]. To realize this vision, the concept of space computing has emerged, which integrates computational power directly within LEO satellites to enable general computing services in space. Typically, LEO satellites used for edge computing applications are deployed at altitudes between 500 and 2000 km, with constellation sizes ranging from a few dozen to several thousand satellites, depending on the specific application scenario. A representative application of space computing is remote sensing [4]. Traditionally, remote sensing images captured by satellites are transmitted back to ground stations for centralized analysis. However, due to the limited bandwidth of satellite-to-ground links and the short satellite transit window, a large portion of the raw data cannot be downloaded to the ground station promptly [5]. For example, during the Kuafu Mission, less than 5% of the monitoring data could be successfully transmitted to Earth [6]. These challenges motivate the idea of space computing by equipping satellites with onboard computing capabilities. By integrating space computing with artificial intelligence (AI), satellites can analyze raw sensing data directly in orbit, automatically identify images of interest, and transmit only the most relevant results back to ground stations. This approach dramatically improves transmission efficiency, reduces end-to-end processing delays, and alleviates the communication burden between satellites and ground stations.

Space computing based on LEO constellation is a new type of computing infrastructure that is fundamentally different from cloud computing and edge computing on the ground [7], [8]. The challenges of space computing come from the inherent characteristics of satellite constellations, such as highly dynamic topology, constrained resources, and harsh space environments [9]. As a paradigm that tightly integrates

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networking, computing, perception, and storage, space computing is increasingly attracting attention from both academia and industry. Yet, most existing work remains limited to theoretical analysis or simulation-based studies, with only a handful of small-scale experimental constellations reported to date [4], [10], [11]. These early implementation trials, however, do not support intersatellite cooperation, thereby constraining their ability to perform constellation-scale computing.

In this article, we systematically present our efforts in the design and implementation of an open space computing platform deployed on a LEO satellite constellation. The initial deployment consists of 12 interconnected satellites, each equipped with intersatellite laser communications and in-orbit AI processing capabilities, enabling distributed, collaborative computation in space. To the best of our knowledge, this represents the largest computing constellation launched to date, providing a unique opportunity to advance research and development in space-based computing. The main contributions of this article are summarized as follows.

- 1) We identify the key enabling technologies essential for space computing, summarize recent advancements, describe our practical implementations, and discuss the remaining challenges. Particular attention is given to spaceborne computers, intersatellite laser communications, spaceborne routers, distributed operating systems, and onboard AI models.
- 2) We design and implement the 3-body computing constellation, an open space computing platform comprising 12 interconnected satellites that support in-orbit distributed computing. We further present in-orbit experimental results, demonstrating that multisatellite distributed computing achieves substantially lower response time than traditional single-satellite workflows.
- 3) We outline several promising research directions that could shape the future evolution of space computing.

The rest of the article is organized as follows. Related work is presented in Section II. The concept and system architecture of space computing are introduced in Section III. The key enabling technologies are outlined in Section IV. We then describe the implementations of our initial 3-body computing constellation in Section V. Following that, future research topics for space computing are discussed in Section VI. Finally, conclusions are drawn in Section VII.

II. RELATED WORK

The concept of aerial and space computing has garnered significant attention in recent years. A comprehensive overview of integrating satellite communications with mobile edge computing (MEC) to support 6G Internet of Things (IoT) applications is provided in [12], particularly in remote or disaster-stricken areas where terrestrial networks are unavailable. MEC extends cloud capabilities to the network edge, enabling low-latency processing near IoT devices. An aerial computing architecture is proposed in [13] to complement existing terrestrial computing paradigms, which is composed of three hierarchical layers: low-altitude computing, high-altitude computing,

and LEO satellite computing. A comprehensive overview to support space-air-ground integrated networks (SAGINs) in infrastructure-less environments is provided in [14], which integrates ground networks, aerial networks, and space networks to enable seamless, high-speed connectivity and data processing for IoT-driven applications. It highlights the importance of computing technologies for serving big data from massive IoT devices.

Driven by advancements in LEO satellite constellations and the increasing demand for real-time data processing in space, space-based computing has emerged as a transformative paradigm to address the limitations of terrestrial infrastructure. An overview of cloud, fog, and edge computing has been presented in [7], affirming that proliferated LEO satellite constellations with optical intersatellite links (ISLs) and onboard processing align more closely with fog computing. Vision and challenges for satellite computing are presented in [15], highlighting a shift from traditional “bent-pipe” architectures, where satellites act solely as passive relays, toward intelligent and autonomous systems. Concerning a specific application of space computing, article [4] proposed a satellite-based framework for remote sensing that is integrated with in-orbit computing and AI services. It highlights benefits like reduced latency by enabling real-time data processing in space, avoiding delays from ground transmissions.

While the above works outline the overall potential of space computing, subsequent studies have focused on different building blocks. With respect to networking, the inherent dynamics of LEO satellites pose many challenges at different layers, such as laser communications at the physical layer, routing at the network layer, congestion control at the transport layer, and task scheduling at the application layer. The problem of network topology design for LEO mega-constellation is investigated in [16], aimed at tackling the high density and high temporal dynamism inherent to the LEO satellites. The opportunities and challenges of LEO mega-constellations, ranging from topology design to routing and congestion control, are discussed in [17]. A framework for simulating and visualizing the network behavior of LEO constellations by incorporating unique characteristics, such as high-velocity orbital motion, is presented in [18]. The cost-performance tradeoffs in the design of network routing that incorporates highly dynamic and time-varying characteristics of satellite connectivity have been examined in [19] and [20], showing that ISLs can substantially reduce the temporal variations in latency and add greater resilience. With respect to satellite edge computing, an in-orbit edge computing architecture based on a nanosatellite constellation and a parallelized pipeline-processing mechanism for remote sensing images is proposed in [21]. Bhattacharjee et al. [3] analyzed the opportunities and challenges of in-orbit computing qualitatively and quantitatively and call for more in-depth study on satellite edge computing from the research community.

Although there are numerous studies trying to address specific issues, the majority of existing results are derived from simulations, which often fail to accurately capture the unique characteristics and constraints of real satellite environments. To bridge the gap between theoretical studies and

practical implementations, several experimental constellations have been launched to facilitate in-orbit experimentation. For instance, a LEO satellite for intelligent remote sensing and in-orbit computing was developed in [4], while the Tiansuan constellation was launched as a research platform to support edge computing on satellites [10], [11], initially comprising six satellites. However, intersatellite communication is not supported in these trials, thereby restricting computing functions within individual satellite rather than enabling constellation-level collaboration. In this work, an initial constellation of 12 computing satellites equipped with intersatellite communication capabilities was launched to enable distributed in-orbit computing, representing the largest space computing constellation to date.

Key challenges remain concerning the stability of intersatellite communications, the design of efficient routing algorithms capable of adapting to highly dynamic topologies, the enhancement of high-performance and reliable onboard computing in harsh space environments, the distribution of workloads across the constellation, and the integration of AI-driven autonomy in orbit.

III. SYSTEM ARCHITECTURE

A. Space Computing Concept

Space computing refers to the deployment and execution of computational tasks directly aboard satellites. It marks a transformative shift from the traditional model where data is acquired in space and processed on the ground, to an integrated architecture in which data is collected and processed entirely in orbit. By pushing computing capabilities into space, this paradigm enables a wide range of aerospace applications, including real-time Earth observation for geological analysis, extraterrestrial asteroid threat monitoring, and enhancements to satellite navigation systems [21].

Compared to traditional terrestrial computing, which relies on ground-based data centers with stable power supplies, abundant computational resources, and fixed network infrastructure, space computing operates in a highly constrained and harsh environment, where limited power, radiation exposure, and dynamic network topologies pose significant challenges. However, space computing offers distinct advantages that terrestrial systems cannot match. First, it allows for the in-orbit processing of space-native data, significantly reducing the volume of data that needs to be transmitted to Earth while also enhancing mission responsiveness. This is particularly beneficial for remote sensing satellites, which generate hundreds of gigabytes of data each day and require continuous, real-time processing. Second, space computing enables truly global coverage for both communication and computation services, even in remote areas or in scenarios where the ground infrastructure is damaged or unavailable [22]. By integrating advanced computing and networking technologies with the latest advancements in AI, space computing is envisioned as a key enabler of the space-ground integrated computing architecture in the coming era of 6G [23].

The computing capability of a single satellite is relatively limited compared to ground-based computing nodes, due

to constraints on size, weight, power, and cost (SWaP-C). To overcome these limitations, the concept of computing constellation is proposed, where multiple satellites within the same orbital plane or across different planes are interconnected to collaboratively execute computational tasks. By distributing workloads across the constellation, this architecture scales beyond the limitations of any individual satellite and supports complex, data-intensive deep-space applications, such as galaxy research and deep-space exploration, which demand high-performance computing capabilities.

B. Space Computing Architecture

Fig. 1 illustrates the architecture of an example space computing constellation, which includes both spaceborne and terrestrial components. The space segment consists of satellites deployed across different orbital planes at altitudes ranging from 500 to 1000 km to ensure continuous coverage. These satellites are interconnected via high-speed intersatellite laser links, forming a resilient mesh network in orbit. The terrestrial segment includes ground stations, mission control centers, and centralized computing and management facilities.

Each satellite is equipped with onboard computing and intersatellite communication payloads, including spaceborne computers, laser communication units, and a spaceborne router. The spaceborne computer serves as the primary computational unit, enabling onboard processing, analysis, and storage of massive volumes of space-native data. The effectiveness of distributed computing is highly dependent on intersatellite connectivity. Satellites communicate with neighbors via ISLs, which can be established using either radio frequency (RF) or laser technologies. In the proposed space computing architecture, laser communication is applied for establishing high-speed ISLs. Typically, each satellite is equipped with four laser units: two are used to establish links with neighboring satellites within the same orbital plane, forming intraplane ISLs, while the other two connect with satellites in adjacent orbital planes, forming interplane ISLs. The spaceborne router is responsible for forwarding data packets from the source to the destination by determining the optimal routing path through the satellite network.

The space computing constellation can be envisioned as a dynamic, distributed computing infrastructure in orbit. By leveraging intersatellite connectivity, reliance on Earth-based infrastructure can be reduced, while a collaborative, space-based distributed operating system can be deployed for efficient task scheduling and resource allocation across satellites according to their workloads and capabilities, extending cloud computing capabilities from Earth into space.

Harnessing the power of space computing enables AI models specifically tailored for space to be deployed directly on satellites, allowing for onboard processing of space-native data and eliminating the need for transmission to ground stations. A key application of this space-based AI capability is image processing, which plays a crucial role in a wide range of space operations, including remote sensing, environmental monitoring, and resource exploration.

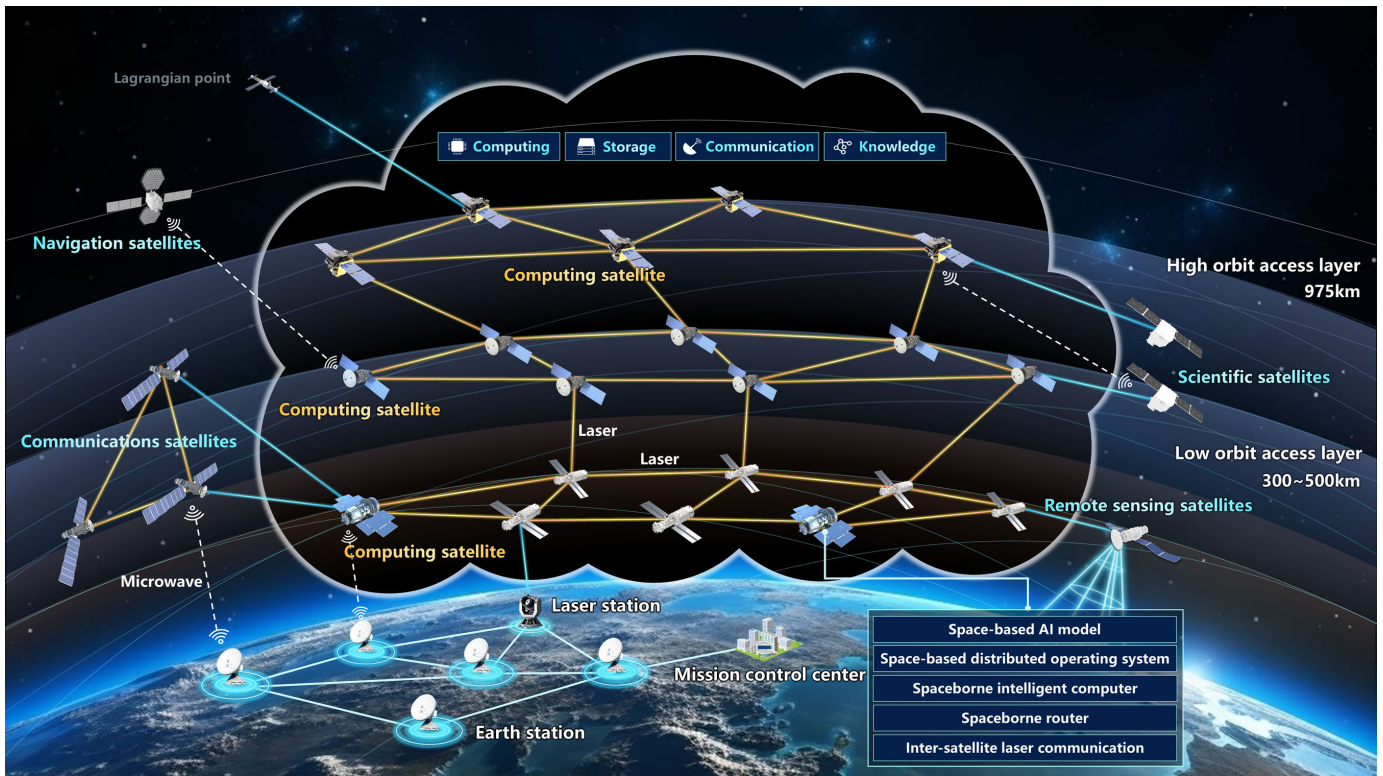


Fig. 1. System architecture of space computing.

IV. ENABLING TECHNOLOGIES

In this section, we identify the key enabling technologies essential for space computing, including spaceborne computers for in-orbit data processing, intersatellite laser communications for high-speed data transfer, spaceborne routers for reliable networking, a distributed operating system for multisatellite coordination, and space-based AI models for onboard inference and autonomous decision-making. For each technology, we review recent advances, describe their practical implementations, and discuss the associated limitations and challenges.

A. Spaceborne Intelligent Computer

1) *State of the Art*: The onboard computer serves as the core of the satellite computing platform, operating under strict constraints of low mass, compact size, and minimal power consumption. It is responsible for generating and encoding telemetry data, receiving, authenticating, and decoding telecommands sent from ground stations, as well as collecting, processing, and storing payload data. Its performance is evaluated based on its processing capabilities, operational speed, and energy efficiency [24]. Enhancing the performance of the onboard computer has become a key area of research and development.

Spaceborne intelligent computers can be classified into general-purpose processors, specialized intelligent processors, and hybrid architectures, based on their processor architecture. Common general-purpose processors include central processing units (CPUs), field-programmable gate arrays (FPGAs), and graphics processing units (GPUs). CPUs offer high

flexibility and generality, supporting multithreading and multitasking. However, CPUs have limited floating-point computing units, making them less suitable for compute-intensive tasks such as deep neural networks and other data-driven intelligent algorithms. FPGAs enable the design of application-specific hardware architectures that can efficiently exploit parallelism. For data-intensive tasks that benefit from parallelism and hardware acceleration, such as image processing and computer vision, FPGAs demonstrate superior energy efficiency compared to traditional hardware accelerators like multicore CPUs and GPUs [25]. Their parallel processing capabilities and low power consumption make them particularly attractive for complex, resource-constrained space computing applications. However, developing sophisticated algorithms on FPGAs is challenging and requires substantial expertise in hardware design. GPUs are equipped with a large number of floating-point units optimized for massive parallel computation. They are also supported by open-source frameworks such as PyTorch and TensorFlow, enabling efficient deployment of neural network inference. The main drawback of GPUs is their high power consumption and the difficulty of heat dissipation in the vacuum of space, which poses practical limitations for satellite deployment.

Specialized intelligent processors are dedicated hardware accelerators designed specifically for AI and machine learning (ML) workloads. Unlike general-purpose processors, they are optimized for high-performance, energy-efficient execution of neural networks and other AI algorithms. Google developed the tensor processing unit (TPU), tailored for TensorFlow, as well as the Edge TPU, a low-power, compact processor

designed for AI inference on edge devices [26]. Microsoft introduced Brainwave, an ultralow-latency deep learning system built on FPGAs [27]. Amazon's Inferentia chip supports a range of mainstream ML frameworks, including TensorFlow, PyTorch, and ONNX [28]. IBM also developed TrueNorth, a neuromorphic processor inspired by brain-like computing architectures [29]. However, most of these specialized processors are not inherently designed for spaceborne environments. Additional radiation hardening and environmental adaptations are required to withstand the extreme conditions of space.

Hybrid architectures, which integrate heterogeneous processing units such as CPUs, GPUs, neural processing units (NPU), and FPGAs on the same chip, module, or board, have emerged as a promising solution to significantly enhance onboard computing performance in terms of processing speed and capability. System-on-chip (SoC) designs are the most prominent examples of these hybrid architectures, combining processor cores, memory units, peripherals, FPGA, and/or GPU fabrics into a single integrated circuit. This integration offers the benefits of reduced power consumption, size, and weight, while simultaneously improving processing capabilities. Most state-of-the-art onboard computers are built on hybrid architectures, particularly CPU + FPGA SoCs [30], [31], and CPU + FPGA + GPU SoCs [32]. The key advantage of using hybrid architectures is the ability to assign specific applications and algorithms to the processing unit best suited for them, thereby achieving optimal performance tailored to specific tasks.

Since onboard computers are intended to operate in extremely harsh space environments, space radiation, characterized by a diverse range of particles, energies, and fluxes, can damage or cause malfunctions in electronic devices and systems, leading to what are known as radiation-induced effects [33]. These effects are broadly categorized into cumulative effects and single-event effects (SEEs). Cumulative effects involve gradual changes in the operational parameters of devices, while SEEs result in abrupt alterations or transient behavior in circuits. Both types of effects can lead to the failure of electronic components, thereby significantly compromising the reliability of space systems.

The most fundamental countermeasure to mitigate the effects of space radiation involves the use of proper packaging and shielding. However, this method increases the overall weight of onboard components and often proves inadequate in fully protecting against ionization phenomena [34]. Radiation-hardened devices, which are specifically designed to withstand radiation-induced effects within defined limits, are commonly used in large satellites. However, their application in LEO satellites is less frequent due to concerns related to energy consumption, cost constraints, and mission lifetime [35]. LEO satellites often rely on commercial off-the-shelf (COTS) components due to their affordability and wide availability, but are more susceptible to radiation compared to space-grade alternatives [36], [37]. Some of the most widely used mitigation techniques include hardware redundancy, firmware protection, and the integration of protection circuits.

Hardware redundancy is an effective strategy for mitigating cumulative effects [36] and single event upsets

(SEUs) [34]. This approach involves integrating duplicate devices or components within the system that perform identical functions. Redundancy comes in two forms: static and dynamic. Dynamic redundancy, also referred to as cold redundancy, is a system-level fault-tolerant technique designed to address cumulative effects. It achieves this by activating a dormant backup component when the primary component becomes unreliable or fails entirely. In contrast, static redundancy, known as hot redundancy, operates redundant components, typically two or three, simultaneously to manage errors induced by SEUs [38].

Firmware is also vulnerable to failures caused by SEUs. Two primary protective strategies are firmware replication [39] and in-orbit firmware updating [40]. Firmware replication involves storing multiple copies of the firmware in different memory locations or across distinct memory devices, ensuring protection against corruption of the primary version [39]. In-orbit firmware updating, on the other hand, utilizes a boot-loader to upload a new firmware image via the uplink channel, enabling direct firmware upgrades while the satellite is in orbit [40]. This technique allows for solving software issues and introducing new features after the satellite has been launched. With the widespread adoption of multicore processors, new fault-tolerance software techniques are emerging, such as task rescheduling or swapping [41].

Protection circuits, such as watchdog timers and overcurrent protection circuits, are essential for safeguarding against SEUs and single-event latchups (SELs). If a processor jumps to an incorrect memory location due to an SEU, the watchdog timer resets the processor to a previous state using a checkpoint, thereby restoring normal operations [42]. Similarly, overcurrent and overvoltage protection circuits monitor for excessive current or voltage levels, triggering a power supply reset when detected to prevent device malfunction caused by SELs [43].

2) Implementations and Challenges: In the implementation of the spaceborne computer, a hybrid architecture that integrates CPU, GPU, and FPGA on the same board has been adopted to provide flexible and efficient computational capabilities for diverse onboard workloads. COTS components are selected as they offer an attractive tradeoff between SWaP, processing performance, and flexibility, but are not inherently radiation-hardened. To enhance system reliability and fault tolerance, high-performance COTS SoCs are integrated with a radiation-hardened microcontroller unit (MCU) in a hybrid architecture. In this design, the COTS SoC serves as a high-performance computing node for computationally demanding tasks, while the radiation-hardened MCU acts as a reliable control node and external supervisor responsible for system monitoring, configuration, and fault management. To further enhance reliability, a multilevel fault-tolerant system has been implemented, featuring error-correcting code (ECC) protected memory and latch-up detection with current-limiting circuits to mitigate radiation effects, watchdog timers, and secure-boot verification to detect system hangs and prevent corrupted firmware from loading, as well as hardware redundancy at both the board and component levels. Additional resilience is achieved through built-in self-test routines and autonomous

recovery protocols, ensuring over 98% availability under harsh space conditions.

In practice, design tradeoffs are evident, as the pursuit of high computational performance (e.g., supporting onboard AI inference for remote sensing) necessitates a balance against stringent power constraints, typically limited to tens of watts per satellite to accommodate solar energy variability and thermal management constraints. Similarly, hardware redundancy strengthens fault tolerance but introduces penalties in mass and volume, thereby restricting overall payload capacity. The current implementation, therefore, emphasizes a balanced design, delivering sufficient onboard computing capability while acknowledging limitations in energy efficiency and vulnerability to radiation-induced degradation. Challenges include further enhancing the performance and reliability of onboard computing under tight power constraints and achieving more seamless integration of AI models with the underlying hardware platform.

B. Intersatellite Laser Communication

1) *State of the Art*: Compared to traditional microwave-based communication, laser communication offers significant advantages for ISLs, including larger bandwidth, smaller size, lighter weight, lower power consumption, stronger resistance to interference, and better security [44]. However, laser links also face significant stability challenges due to high-speed orbital movements, making rapid beam acquisition and high-precision beam tracking critical for maintaining reliable communications. This relies on advanced optical tracking and aiming systems to ensure dynamic alignment and connection between satellites, while factors such as thermal fluctuations, mechanical vibrations, and orbit prediction error can degrade system performance [45]. Additionally, environmental disturbances such as background noise from solar, lunar, and planetary radiations can sometimes exceed the intensity of the received signal, demanding enhanced noise resistance and signal-processing capabilities [46], [47]. Overcoming these challenges is critical to establish high-speed and reliable ISLs between satellites.

The successful implementation of intersatellite laser communications relies on several advanced technologies, including optical transceiver design, acquisition and tracking (A/T), and error correction techniques. Optical transceivers can be categorized into direct detection and coherent detection. Direct detection has a simpler architecture but lower sensitivity. Coherent detection, on the other hand, requires additional modules such as local oscillator lasers, optical mixers, and electrooptic modulators, resulting in a more complex architecture and higher device requirements, but it offers significantly higher sensitivity [48]. Theoretically, coherent detection can improve receiving sensitivity by 1–2 orders of magnitude compared to direct detection. Additionally, coherent detection can be combined with high-order modulation and wavelength-division multiplexing (WDM) techniques, further enhancing transmission data rates. A comparison between the two optical transceiver designs is shown in Table I. Direct detection is well-suited for application scenarios with relatively short link distances and lower data rates, whereas coherent detection

TABLE I
COMPARISON BETWEEN DIRECT DETECTION AND COHERENT DETECTION

KPI	Direct detection	Coherent detection
Receiving sensitivity	~ 20 dB lower than coherent detection	High, with advantages of reduced size and power
Anti-interference	Necessary to avoid interference from spatial background light	Resistance to background noise, immunity to solar conjunction interference
System complexity	Simple receiver structure, easy to implement	Complex receiver structure, difficult to implement
Cost	Off-the-shelf products, low cost	Need to add frequency mixer, phase shifter, and local oscillator, high cost

is particularly suitable for long-distance, high-rate transmission and is, therefore, a primary focus for intersatellite communications.

Acquisition, pointing, and tracking (APT) systems are critical for establishing and maintaining precise alignment of laser beams between fast-moving satellites. The APT system consists of several key components. Gimbal systems, typically using two-axis or three-axis gimbals, allow the optical payloads (e.g., laser transmitters and receivers) to be dynamically adjusted in real time to follow the satellite's position and maintain a line-of-sight (LOS) between satellites. Fine pointing systems are essential to ensure that the laser beam remains focused on the target satellite even as both satellites move at high speeds. This is typically achieved through closed-loop control systems that continuously adjust the position of the laser beam, which rely on high-precision sensors and actuators to make microadjustments in response to small deviations. Inertial measurement units, including gyroscopes and accelerometers, monitor satellite motion and vibrations to aid in beam stabilization. Optical tracking sensors are used to detect and lock onto the laser beam emitted by the satellite's communication payload. The A/T subsystem is used to initially acquire the signal when the satellites come within LOS and then track it as the relative positions of the satellites change. Once the laser link is established, the A/T subsystem continuously adjusts the satellite's position to ensure the beam stays focused on the receiving satellite. Sophisticated orbit prediction algorithms are used to calculate the relative motion between satellites, predicting their positions and orientations. This allows for proactive adjustments in the pointing systems, helping to anticipate and correct any misalignment before it leads to a communication link failure.

2) *Implementations and Challenges*: In the implementation of intersatellite laser communication, high-speed optical terminals that integrate precision APT mechanisms with coherent detection are employed. The terminal adopts a dual-stage APT system with automatic beam A/T, consisting of coarse pointing using gimbals and star sensors, followed by fine pointing through fast steering mirrors and closed-loop control to maintain precise alignment under platform jitter and dynamic orbital motion. The combination of coherent detection and advanced digital signal processing significantly enhances receiver sensitivity and spectral efficiency, enabling gigabit communications while mitigating the effects of background noise and Doppler-induced frequency shifts.

To improve link resilience, a conventional RF transceiver is retained as a fallback channel in the event of optical link interruption. Additionally, radiation-hardened detectors and ECCs are incorporated to mitigate bit flips and burst errors in optical communication electronics.

Despite recent progress, intersatellite laser communication still faces substantial challenges in maintaining precise beam alignment between fast-moving satellites. Current systems also experience long link acquisition times and limited link stability. Overcoming these limitations requires advances in high-precision real-time attitude and orbit data sharing to support accurate pointing prediction and coordination among satellites, as well as high-speed, high-accuracy beam scanning and tracking algorithms to shorten acquisition time and maintain alignment during orbital maneuvers. Emerging intelligent link prediction and prewarning technologies, which leverage AI and predictive modeling, also offer promising means to anticipate link interruptions and dynamically optimize communication parameters. These advancements are essential to enhance stability and reliability, enabling intersatellite laser networks to serve as a robust, high-speed optical backbone for distributed in-orbit computing.

C. Spaceborne Router

1) *State of the Art*: Routing is a fundamental function of communication networks to ensure reliable transmission paths. Although existing routing protocols such as open shortest path first (OSPF) and border gateway protocol (BGP) have demonstrated effectiveness in terrestrial networks, they cannot be directly applied to satellite networks because of the inherent dynamics of LEO satellites. Consequently, the design of routing algorithms for LEO satellite networks must satisfy the following five key requirements: **adaptability**, to cope with rapidly changing topologies caused by satellite motion across different orbits; **efficiency**, to minimize routing overhead under constraints of limited energy, processing, and storage resources; **resilience**, to quickly detect and recover from disruptions caused by the harsh space environment such as malfunctions, interference, or collisions; **load balancing**, to distribute traffic evenly across satellites and links, avoiding congestion and performance degradation; and **scalability**, to maintain robust performance as the constellation grows in size across multiple orbital planes.

Most existing LEO network routing algorithms can be classified into proactive and reactive routing [49], [50], [51]. A proactive routing predicts the topology change and computes routing paths for each snapshot of the network topology. The duration of each snapshot is shorter when the topology changes more frequently, requiring more snapshots to be calculated. This results in increased overhead due to the need for more frequent updates. On the other hand, a reactive routing protocol detects topology changes in real time and advertises the changes to all routers or to a centralized controller. The overhead of such operations becomes very high if the topology changes frequently. The intrinsic reason for such inefficiency is that the advertisements and routing recalculations need to be performed on a global scale. However, as demonstrated by many studies, some topology changes have limited or

no impact on the routing of certain parts of satellites. This suggests that global topology awareness may not be necessary.

One possible solution to improve the routing efficiency and scalability while maintaining resilience to link or node failures is hierarchical routing [52]. The idea is to divide the LEO constellation into areas and divide the routing into intraarea routing and interarea routing. The areas should be carefully designed in such a way that the impact of link failure to interarea routing is minimized, and only the intraarea routing is affected. Furthermore, the size of each area should be properly configured so that intraarea routing can be reconstructed quickly with minimal overhead. Link-state routing protocols such as OSPF can be used for intraarea and interarea routings to handle topology changes caused by regular satellite movements or unpredictable link/node failures.

2) *Implementations and Challenges*: The spaceborne router is built upon an aerospace-grade, high-performance network switching chipset serving as its hardware foundation. The system software is based on the open-source network operating system SONiC, which employs a containerized architecture to enable modular management and high flexibility. Core routing functions are implemented through the free range routing (FRR) protocol container, while application modules for telemetry and telecommand, watchdog monitoring, single-event detection, and configuration deployment operate atop SONiC. This layered and modular architecture provides flexible and reliable in-orbit networking capabilities.

The implementation of the routing algorithm leverages the periodicity and regularity of satellite movements to efficiently handle predictable topology dynamics. To exploit this property, the routing framework is structured into three key functional stages: route calculation, route injection, and time-slice switching. In the route calculation stage, the network controller divides the constellation's orbital period into a sequence of static topology snapshots based on the ephemeris. For each snapshot, the controller precomputes the static topology database, neighbor table, and forwarding table based on orbital parameters and constellation geometry. These precomputed routing tables are then disseminated to onboard routers during the route injection phase, ensuring synchronization across the constellation. Finally, time-slice switching allows satellites to transition between routing tables according to orbital motion, avoiding frequent recomputation during predictable topology changes.

Nevertheless, LEO satellite networks are also subject to irregular topology variations arising from unexpected link failures, which cannot be efficiently handled by static snapshot-based routing alone. To address this limitation, our spaceborne router integrates dynamic routing elements into the static framework in an on-demand manner. Under normal conditions, the system relies on static snapshots to minimize control overhead and eliminate convergence delays. When irregular events occur, dynamic routing protocols such as OSPF are selectively invoked to perform localized flooding and path recalculation, thereby achieving fast convergence while limiting signaling overhead. This hybrid approach combines the efficiency of precomputed static routing with the adaptability of dynamic routing, striking a balance between

predictable orbital regularity and resilience against unexpected disruptions.

The resilience of the spaceborne router is ensured through a dual redundancy strategy that combines cold-standby routing and link backup. In the cold standby design, a primary router handles real-time forwarding while a backup remains in standby, which will be automatically activated upon hardware faults, software anomalies, or security attacks, at the cost of doubling hardware resources. Complementarily, link redundancy provides multiple physical and logical channels with fast switchover to mitigate port failures or performance degradation. These redundancy mechanisms, together with dynamic routing, enhance fault tolerance while achieving a balance between resilience, cost, and engineering feasibility.

The current implementation has demonstrated promising progress in improving routing efficiency for LEO constellations, but limitations remain. While the hybrid routing framework can manage regular topology changes and occasional disruptions in small-scale constellations, it struggles to achieve fast convergence and stable performance in large-scale LEO constellations. Furthermore, the current design does not fully support integrated satellite-terrestrial networking, limiting seamless end-to-end connectivity between space and ground infrastructures. Future work should focus on developing intelligent, scalable, and interoperable routing architectures that bridge satellite constellations with terrestrial networks.

D. Space-Based Distributed Operating System

1) *State of the Art*: Traditionally, onboard satellite software has been rigidly designed for mission-specific functions, resulting in single-purpose systems that lack generalization and reusability across different platforms [53]. Recent advancements in satellite hardware and software are driving toward more generalized architectures [8]. Cloud-native technologies, characterized by containerized microservices, Kubernetes orchestration, DevOps automation, and continuous delivery, facilitate a more scalable, cost-effective, and rapid deployment for applications in space environments [11]. By decoupling software from hardware, cloud-native architectures enable dynamic updates, in-orbit reconfigurability, and efficient resource utilization. This paradigm allows satellites to run multiple applications concurrently, support AI-driven autonomy, and distribute computational workloads across constellations.

However, the implementation of cloud-native technology to satellites is still a relatively new concept and faces unique challenges, including intermittent connectivity, constrained resources, and the need for fault tolerance in radiation-prone environments. Furthermore, the dynamic topology of satellite constellations adds another layer of complexity in managing heterogeneous resources, synchronizing tasks across nodes, and ensuring autonomous operations without constant ground control, for which traditional cloud systems are not inherently designed to handle. Overcoming these challenges requires rethinking traditional operating system architectures and adapting cloud-native principles to function reliably within the constraints and unpredictability of space.

The cloud-native satellite architecture is illustrated in Fig. 2, which comprises five layers. The physical hardware layer forms the foundation, providing the essential computing, storage, and networking resources required to support higher level virtualized and service-oriented components. In the context of space-based cloud computing, this layer encompasses various satellite payloads, including onboard processors, memory, AI accelerators, communication modules, and sensors. Robust hardware design and fault-tolerant mechanisms at the physical layer are essential to ensure the reliable delivery of cloud services for satellites.

The virtualization layer serves as a critical abstraction in cloud architecture, enabling efficient resource utilization by decoupling software from underlying physical hardware. Through virtualization and container technologies such as virtual machines (VMs), containers, and hypervisors, this layer enables multiple workloads to run concurrently on shared hardware or across distributed satellite nodes, offering scalability, portability, and isolation.

The operating system layer serves as a critical bridge between the underlying physical or virtualized hardware and higher level containerized services and applications. There is an increasing need to develop a generalized operating system that can be seamlessly deployed on satellites [54]. For instance, it should be compatible with the existing Linux software ecosystem, allowing smooth migration of existing software to satellite systems. Additionally, the satellite operating system must be upgradable to accommodate evolving software patches, security updates, and new functionalities without requiring physical access to the satellite [55]. As a platform for space computing, the operating system must also be AI-algorithm friendly to support the integration, deployment, and execution of advanced AI capabilities in space [56]. This includes support for AI-specific libraries and frameworks, hardware acceleration for model inference, efficient memory and process management, and real-time data handling. Furthermore, the operating system must be lightweight and modular to minimize resource consumption while providing essential functions such as device drivers, interprocess communication, and network coordination across satellite constellations.

The service and orchestration layer is responsible for the dynamic coordination, deployment, and lifecycle management of distributed services across heterogeneous resources. Containers and microservices are the key technologies for realizing cloud-native satellite architectures. Lightweight orchestration frameworks, which can be adapted from terrestrial tools like Kubernetes and KubeEdge [57], must be reengineered to manage containers and microservices, where decisions are made locally or within small clusters without reliance on continuous ground-based control. Applications and functions can be deployed on satellites in the form of containers or microservices for flexible development and orchestration. This layer is also critical for coordinating tasks across multiple satellites, facilitating service migration, and enabling adaptive resource allocation in response to changing mission demands or environmental conditions. The problem of intersatellite orchestration has been investigated in [58], where traditional container technologies are deployed across multiple satellites

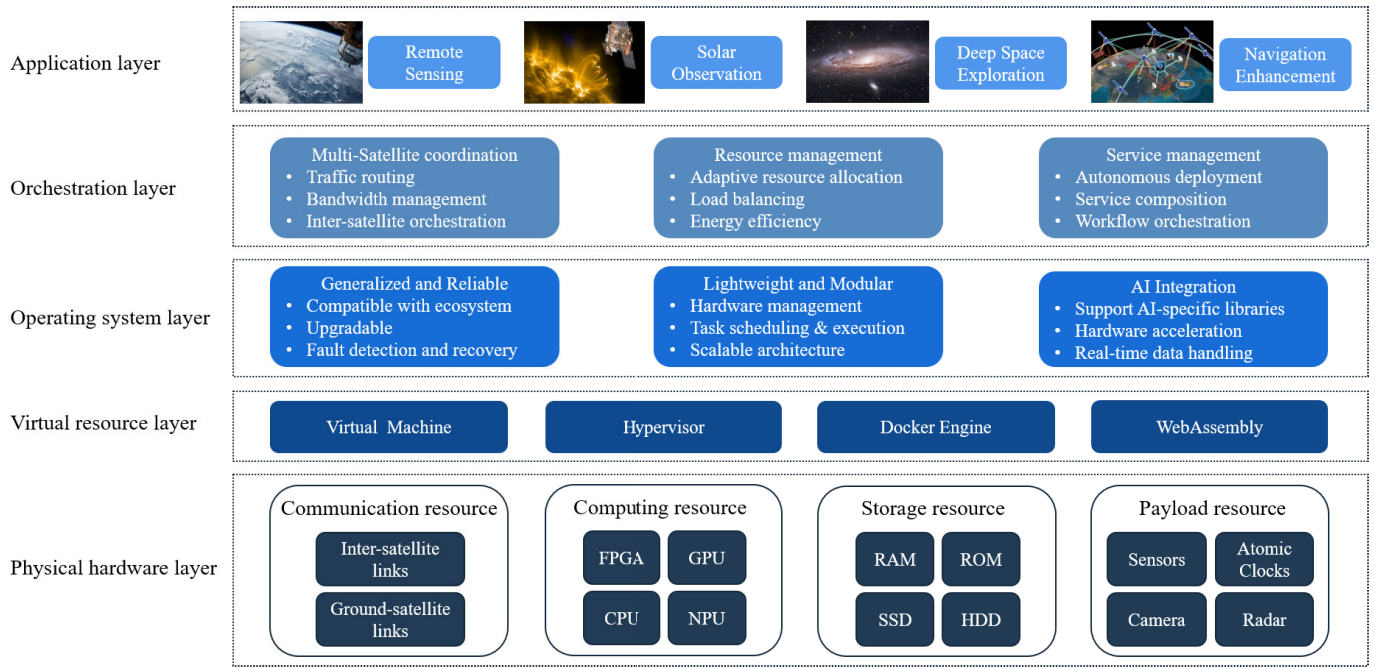


Fig. 2. Cloud-native satellite architecture.

co-located with the ground station. It is also interesting to investigate inner-satellite orchestration, as modern satellites often integrate multiple heterogeneous computing units within a single satellite. With this multilevel intelligent orchestration, the computing efficiency can be significantly enhanced.

The application layer serves as the interface where user applications and intelligent algorithms are deployed and executed. This layer hosts a wide range of space-centric workloads. Unlike traditional static onboard applications, the space-based cloud application layer should be designed to be compatible with containerized service architectures, facilitating dynamic deployment, flexible updates, and seamless cross-node collaboration across satellite constellations. Applications running at this layer can leverage in-orbit AI inference to reduce reliance on ground stations and enhance autonomous operations.

2) *Implementations and Challenges:* The implementation of the space-based distributed operating system adopts a layered architecture spanning both the space and ground segments, as illustrated in Fig. 3. It is composed of four key functional modules: resource pooling, resource management and task orchestration, virtual cluster autonomy, and data collaboration and management. The resource pooling software virtualizes onboard computing units through fine-grained partitioning, aggregation, and reuse, enabling applications to transparently share satellite computing resources within containerized environments. This capability is implemented using Kubernetes together with device plugins, GPU container runtimes, and toolkits, which employ time-slicing to partition a single GPU into multiple virtual instances. Containerized deployment further enables applications and services to be encapsulated into lightweight, isolated units that can be rapidly launched or migrated across satellites with minimal configuration overhead. Resource management and task

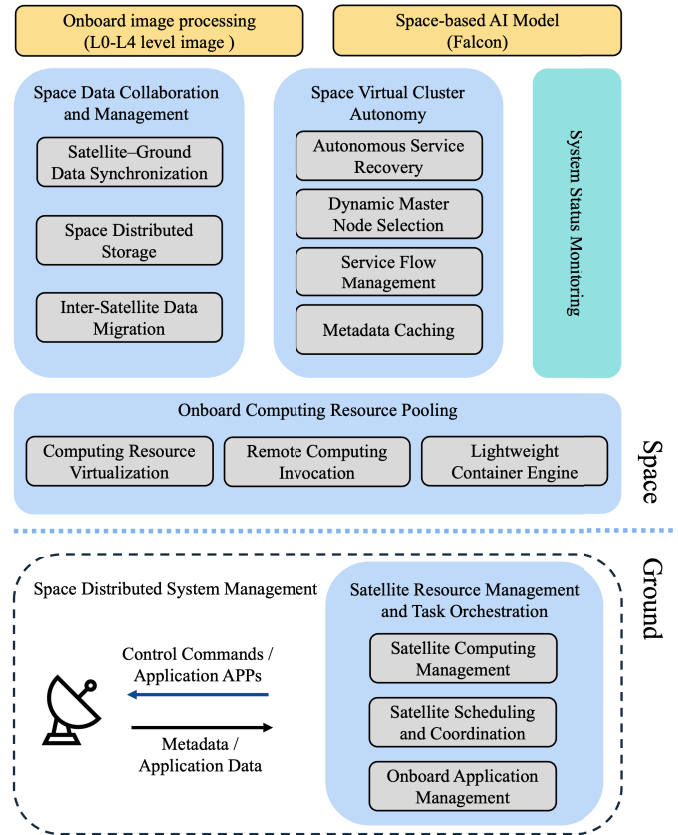


Fig. 3. Implementation of the space-based distributed operating system.

orchestration abstract computing, networking, storage, and payload resources into a unified constellation-wide pool. When a task arrives, the system evaluates orbital geometry, link conditions, onboard workload, resource availability, and data

locality to determine whether execution should occur on a single satellite or be distributed across multiple nodes. Tasks and intermediate results are exchanged through ISLs, and resource usage is continuously monitored so that workloads can be reallocated or migrated in response to congestion, faults, or topology changes. The virtual cluster autonomy software maintains task continuity under intermittent ground connectivity by enabling offline autonomy and live migration of applications from faulty to healthy units. When ground connectivity is available, it updates and synchronizes cluster states to the ground, ensuring uninterrupted operation in the presence of failures. Finally, the data collaboration management software, implemented via a CubeFS-based distributed file system, supports data synchronization across satellites and with the ground, pooling storage resources to provide reliable and efficient data sharing despite short communication windows.

To ensure reliable operation in a harsh space environment while meeting stringent SWaP constraints, the operating system is designed to be lightweight, energy-efficient, and equipped with robust fault-tolerant mechanisms capable of detecting, isolating, and recovering from hardware failures and software anomalies. These mechanisms include container-based process isolation and health-monitoring daemons for real-time detection of abnormal resource or application behavior, checkpointing with automated rollback for fast recovery, cross-satellite service migration, and redundant distributed storage to preserve data integrity. Additionally, the operating system maintains compatibility with the Linux ecosystem to facilitate seamless updates, patch management, and rapid application deployment. This compatibility not only reduces the development and integration cycle but also enhances flexibility in adopting emerging tools and frameworks.

Despite these capabilities, several design challenges and tradeoffs remain. Virtualization and containerization provide portability and flexible orchestration, but introduce additional CPU and memory overhead that may degrade performance under limited onboard resources. Techniques such as live migration and distributed storage pooling improve resilience and fault tolerance, but increase system complexity, bandwidth consumption, and processing load. Moreover, synchronization across intermittent and delay-prone satellite-ground links introduces latency and risks of inconsistency, requiring efficient coordination and conflict-resolution mechanisms to maintain a coherent global state.

E. Space-Based Vision–Language Foundation Model

1) *State of the Art*: There is growing interest in extending AI into space by deploying large models on satellites to enhance onboard data processing. By leveraging the computational power of large models onboard, real-time processing of remote sensing data becomes possible, thereby reducing the need for extensive data transmission. While large vision–language models (LVLMs) have demonstrated strong performance on natural images, developing a remote sensing foundation model capable of multilevel understanding and reasoning across images, regions, and pixels remains a major challenge.

The performance of LVLMs largely depends on the quality and quantity of training data and the fine-tuning techniques used. The development of high-quality remote sensing datasets has attracted increasing attention in recent years. Studies in [59], [60], and [61] proposed various image datasets, each aimed at a single or a few vision tasks such as object detection, scene classification, and image segmentation. Besides, studies in [62], [63], and [64] are focused on developing multimodal datasets, which include remote sensing images, visual reasoning, and conversational tasks, to satisfy the needs of vision–language models. However, existing remote sensing datasets still remain limited in terms of data scale, task diversity, hierarchical annotation, and annotation quality [15]. It is crucial to develop a large-scale, high-quality, multitask remote sensing dataset for training foundational vision–language models. One straightforward approach is to curate and combine high-quality training data from multiple open sources.

Integrating data from diverse sources is of great importance to train LVLMs comprehensively. However, the heterogeneous nature of these diverse data sources introduces significant technical challenges in integration, such as inconsistent annotation formats and category naming, necessitating effective data preprocessing and compatibility strategies. Data cleaning, normalization, feature engineering, and multimodal data alignment are commonly used preprocessing techniques to effectively transform raw data into compatible formats, ensuring that data from various sources is standardized. Besides, ensuring data privacy and security is also an important issue in data integration.

Selecting and fine-tuning appropriate foundation model architectures and pretrained models is another critical consideration. The pretrained foundation models can be categorized based on architectural design. The first category is ViT-based vision foundation models [65], [66], which lack reasoning capabilities, thus cannot be controlled via natural language instructions. The second category includes CLIP-based vision–language models, of which RemoteCLIP [67] is the first vision–language foundation model for remote sensing that aligned text embeddings. These models perform well in specific remote sensing tasks but cannot adapt to more diverse and complex scenarios for handling multiple tasks, due to the lack of comprehensive perceptual and reasoning abilities. The third category consists of large language models (LLMs)-based vision–language models, such as GroChat [68], a versatile LLaVA-based remote sensing vision–language model, and LHRS-Bot [63], a VGI-enhanced large multimodal language model for remote sensing. These models often exceed 7 billion parameters, leading to high computation, memory, and energy costs when deployed on satellites, which have limited computing and energy capabilities.

2) *Implementations and Challenges*: During the implementation of the space-based AI model, it is observed that an extremely large number of parameters typical of LLMs may be unnecessary for remote sensing applications, as most tasks primarily rely on visual inputs. Therefore, a lightweight vision–language foundation model, named Falcon, is developed to satisfy the stringent resource constraints [69]. In particular, Falcon features a unified architecture for

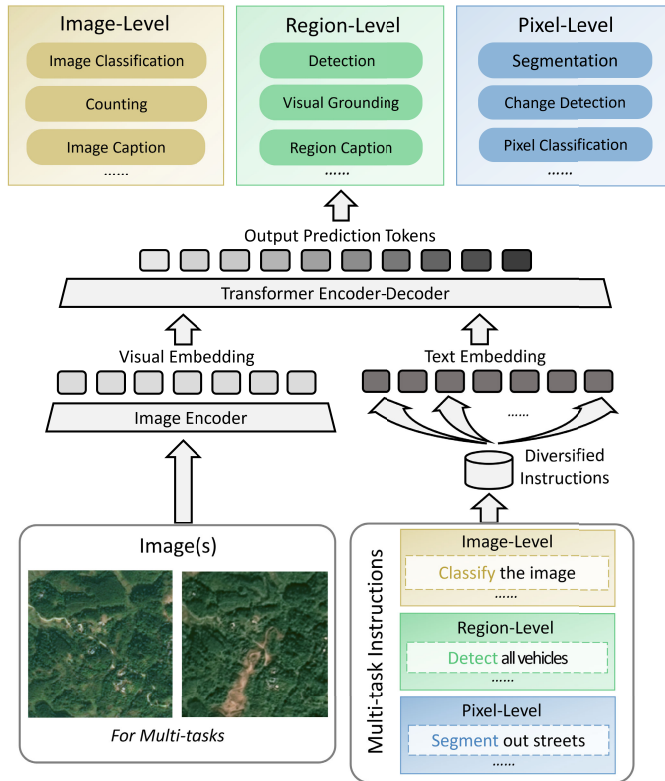


Fig. 4. Overview of Falcon model architecture.

multitask learning, bridging image-level, region-level, and pixel-level reasoning and understanding abilities in one model. The main challenge for designing Falcon is to learn a universal representation for diverse vision–language tasks. Inspired by the latest research in the natural image area [70], [71], a unified network architecture to seamlessly integrate spatial hierarchy and semantic granularity information into a universal representation is proposed, which consists of an image encoder and a transformer-based multimodality encoder–decoder [69].

Specifically, Falcon adopts a sequence-to-sequence pipeline that supports a wide range of remote sensing tasks within a unified model architecture, as depicted in Fig. 4. Given a remote sensing image and a natural-language instruction, the image is first processed by a visual encoder to produce visual feature tokens, while the instruction is converted into text embeddings by a language encoder. These two streams are aligned through a visual adapter, which maps image features into a shared latent space and concatenates them with the instruction representation to form a single multimodal input sequence. This fused representation is then fed into a transformer-based encoder–decoder network, which autoregressively outputs the task results in textual form. In this formulation, all tasks, such as classification, object detection, segmentation, captioning, and change detection, are treated as text generation problems, eliminating the need for separate task-specific heads. Falcon also employs a dynamic prompt training strategy, where multiple paraphrased versions of each instruction are used during training to enhance robustness to diverse user inputs. This unified pipeline enables Falcon to handle various image-level, region-level, and pixel-level tasks without modifying the model structure.

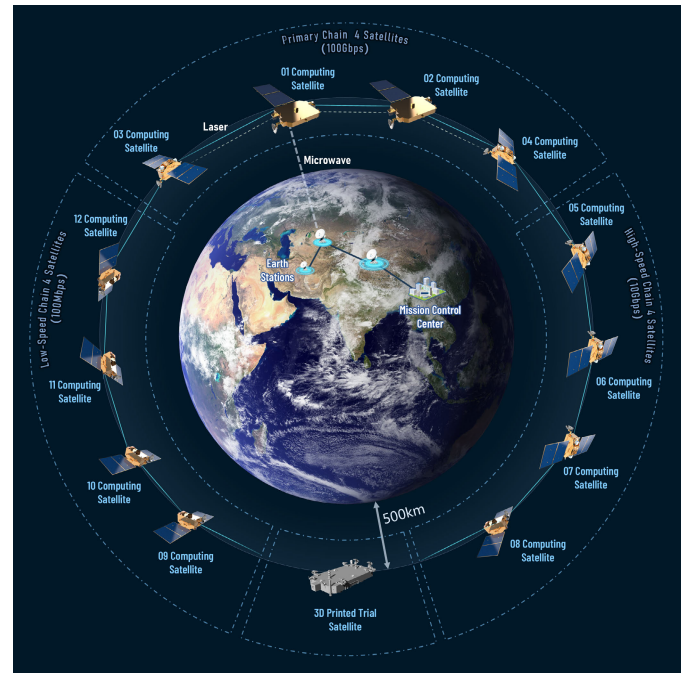


Fig. 5. Initial constellation, which consists of 12 computing satellites in the Sun-synchronous orbit at 535-km height, and one 3-D printed experimental satellite.

The primary design tradeoff for Falcon lies in balancing model complexity against the satellite’s SWaP constraints. Reducing parameter size facilitates onboard deployment and real-time inference, but may increase sensitivity to environmental noise and task variability. Achieving high performance under limited computational and energy budgets, therefore, requires efficient model optimization techniques. In parallel, dataset construction remains a major challenge, as scalable data curation pipelines must integrate multiresolution, multi-modal, and cross-platform sources while keeping annotation costs manageable.

V. 3-BODY COMPUTING CONSTELLATION

In this section, we present our efforts toward the implementation of an open space computing platform, named the 3-body computing constellation, together with ground validations and in-orbit experiments.

A. Overview

The 3-body computing constellation, initiated by Zhejiang Lab, is a large-scale space computing infrastructure built on LEO satellites. It serves as an open platform that supports access to real satellite datasets to facilitate research and experiments directly on the real satellite network. The 3-body computing constellation is envisioned as the backbone for providing advanced computing services to a wide range of existing satellite constellations, including those dedicated to communication, navigation, remote sensing, and space science, empowering them to perform complex, computation-intensive data analysis directly in orbit.

The constellation is planned to be rolled out in three phases: 12 satellites in the initial phase, 100 satellites in the second phase, and 1000 satellites in the third phase. Fig. 5 shows the

topology and configuration of the initial constellation, which consists of 12 computing satellites and one 3-D printed experimental satellite. Among the computing satellites, two serve as primary nodes while the remaining ten function as secondary nodes. The initial constellation has been successfully launched into the 535 km Sun-synchronous orbit with an inclination of 97.5° in May 2025.

As an initial deployment, the 12-satellite configuration strikes a practical balance among latency, coverage, intersatellite connectivity, and system cost. While a single satellite exhibits a revisit time of approximately 90 min, a 12-satellite constellation reduces this to under 10 min and provides frequent intersatellite contact opportunities, enabling cooperative execution and sub-10-min response times, as demonstrated in our in-orbit experiments later.

Each satellite is equipped with two laser communication terminals to establish high-speed ISLs with neighboring satellites within the same orbital plane, supporting data rates of 100 Mb/s, 10 Gb/s, and up to 100 Gb/s. To enhance communication reliability, backup radio transceivers are incorporated to serve as a fallback channel in the event of laser link failure. A custom-designed spaceborne router is integrated into each satellite to ensure efficient and robust connectivity across intrasatellite payloads, ISLs, and satellite-to-ground links. High-performance and reliable computing services are provided by two types of spaceborne computers: NX1 and G1. The NX1 is based on a GPU SoC architecture, while the G1 adopts a hybrid processor architecture integrating FPGA and GPU SoC. Both modules deliver a computational throughput of 248 trillion operations per second (OPS). A high-throughput computing engine was developed to enhance real-time performance for onboard algorithm libraries and large AI models. The spaceborne computer also supports customized configurations through an elastic, scalable hardware–software co-design framework, enabling tailored deployment for different mission requirements within the constellation. Each primary computing satellite is equipped with one NX1 and two G1 modules, achieving a combined 744 TOPS, whereas each secondary computing satellite carries a single NX1 module, supporting 248 TOPS. Collectively, the initial constellation delivers up to 5 peta-OPS of in-orbit computing capacity and offers 30 TB of distributed storage.

Built on a containerized framework, a space-based distributed operating system is deployed across the constellation to enable efficient task scheduling and coordinated management of computing, communication, and storage resources. The system is highly lightweight, featuring an optimized spaceborne operating system of less than 100 MB and typical applications under 2 MB, which significantly reduces onboard resource consumption. It supports flexible deployment and updating of applications, incorporates autonomous virtual constellation management under intermittent connectivity, and provides comprehensive monitoring of application states and computing resources across the network.

For remote sensing applications, pretrained, domain-specific AI models are deployed on satellites to perform in-orbit data analysis. The acquired raw data must first undergo a series of image preprocessing steps onboard to ensure input quality

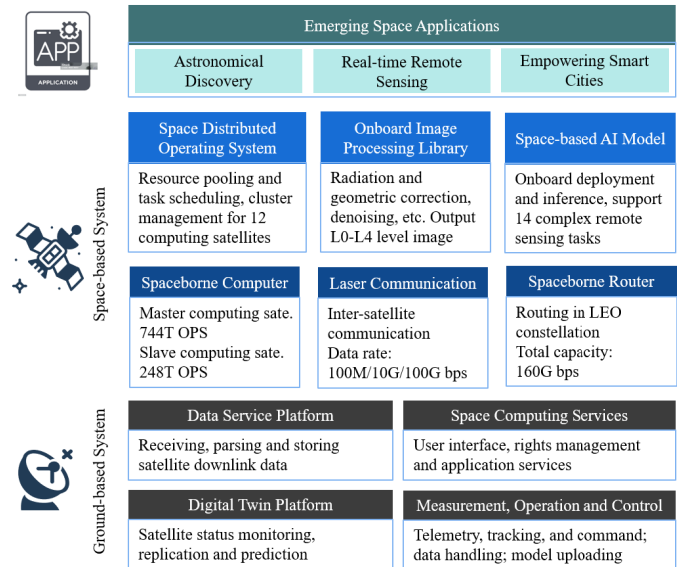


Fig. 6. System diagram of the 3-body computing constellation.

and consistency before it can be fed into the models for inference. To support this, a dedicated remote sensing image processing software package has been developed, providing a comprehensive suite of onboard image-processing functions. These include radiometric correction to mitigate sensor and atmospheric distortions, band alignment to maintain spatial coherence across spectral bands, and geometric correction to accurately align imagery with geographic coordinates. The software also incorporates noise reduction to enhance image clarity, cloud detection to identify and mask cloud-covered regions, and image tiling to partition large datasets into manageable segments. Additional capabilities include image fusion for integrating data from multiple sensors, position calculation for precise geolocation, and image compression to optimize storage and transmission efficiency. Once preprocessing is complete, the refined data is fed into pretrained AI models to perform a variety of onboard inference tasks.

The system architecture and technical specifications of the key components in the initial computing constellation are presented in Fig. 6 and summarized in Table II, respectively. The fault tolerance mechanisms implemented across the hardware, networking, and distributed systems are outlined in Table III. Collectively, these mechanisms provide multilevel resilience in the presence of failures. As this initial deployment is primarily intended for technology validation, continuous 24/7 operation was not a design objective. Instead, payload activation is approximately duty-cycled to maintain thermal and power balance.

B. AI in Space

Modern LEO satellites generate massive volumes of remote sensing imagery, around 100 TB per day, which imposes heavy bandwidth pressure on satellite-to-ground links. Additionally, adverse weather further reduces image quality, straining limited downlink resources. Onboard AI mitigates these issues by prioritizing and transmitting only the most valuable data to the ground.

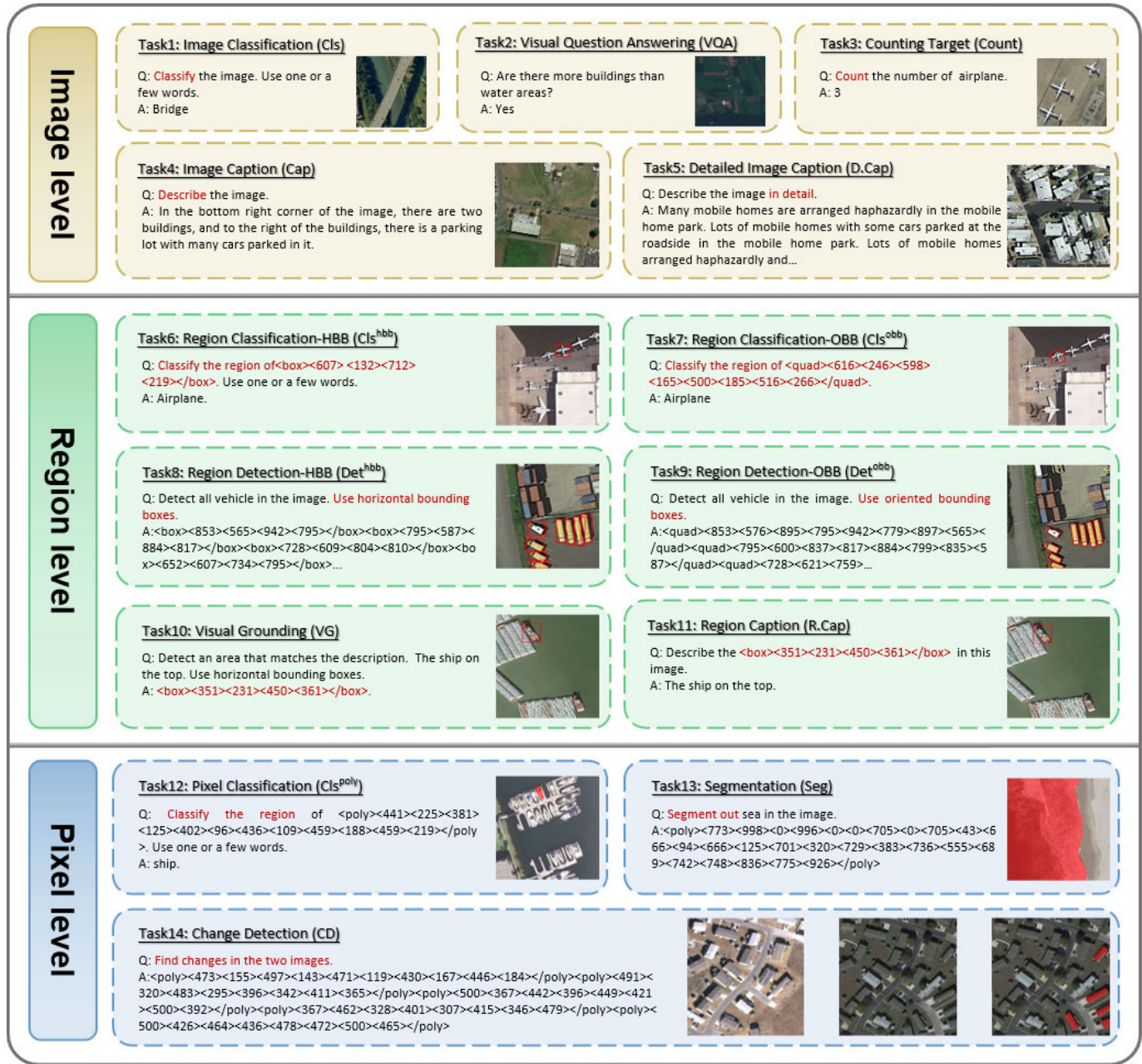


Fig. 7. Illustrative example of images, their corresponding instructions, and output format of supported tasks in Falcon.

Therefore, we developed Falcon, a versatile vision–language foundation model deployed onboard with comprehensive perceptual and reasoning abilities tailored for remote sensing, supporting 14 complex tasks within a unified architecture [69]. The full list of supported tasks is illustrated in Fig. 7. The model is designed to be lightweight, comprising only 0.7 billion parameters, and its weights are initialized from the pretrained model provided in [71]. The output token length is increased to 4096 to obtain more detailed representations. The training batch size for Falcon is set to be 640, the learning rate is set to $1e^{-5}$, and the image size is 448×448 . The model was trained for four days using 160 Nvidia A100 GPUs.

The model parameters are composed of two parts: the base model parameters and the low-rank adaptation (LoRA)

incremental parameters. After the satellite is deployed, researchers can continue to fine-tune and update the onboard model. New knowledge acquired through this process is distilled into the LoRA incremental parameters, allowing the base model to remain unchanged while incorporating recent adaptations. To ensure efficient transmission, these incremental parameters are quantized, reducing the file size to less than 50 MB, small enough to be fully uploaded during a single satellite-to-ground communication session. This approach enables continuous model improvement while minimizing communication overhead and preserving limited bandwidth resources.

To facilitate Falcon’s training, we compiled a comprehensive, large-scale, multitask instruction-tuning dataset for

TABLE II
MAIN TECHNICAL SPECIFICATIONS OF THE 3-BODY COMPUTING
CONSTELLATION IN PHASE ONE

Components	Specifications
Onboard computer NX1	Computing power: 248 TOPS @int8 Architecture: GPU-SoC + FPGA-SoC Memory (GPU): 64 GB + ECC Memory (FPGA): 4 GB(PS) + 4 GB(PL) + ECC Storage: 8 TB Power consumption: 51W (Standby), 128W (Peak) Weight: 2.35 kg Dimension: 231mm × 160mm × 45.5mm
Onboard computer G1	Computing power: 248 TOPS @int8 Architecture: GPU-SoC Memory : 64 GB + ECC Storage: 2 TB Power consumption: 30W (Standby), 83W (Peak) Weight: 1.08 kg Dimension: 210mm × 157.2mm × 30mm
Master computing satellites (01-02)	Computing hardware: NX1*1 + G1*2 Total computing power: 744 TOPS @int8
Slave computing satellites (03-12)	Computing hardware: NX1*1 Total computing power: 248 TOPS @int8
System computing and storage	Total computing power: 5 POPS @int8 Total storage: 30 TB
Spaceborne router	Max rate of a single port: 10 Gbps Overall switching capacity: 160 Gbps Routing protocols: OSPF, BGP Power consumption: 40W (Peak 45W) Forwarding delay: 660 ns Weight: 3 kg
Spaceborne laser terminal	Rate: 100 Mbps, 10 Gbps, 100 Gbps Distance: 500-1500 km Power consumption: 100 Mbps: 20W (Peak 34W) 10 Gbps: 65W (Peak 96W) 100 Gbps: 85W (Peak 112W) Weight: 4.7 kg (100M), 8.2 kg (10G, 100G)
Space-based operating system	Lightweight containerized framework (< 100 MB) User-defined onboard application upload Dynamic orchestration and scheduling Virtual cluster autonomy
Onboard image processing	Functions: Radiation and geometric correction, denoising, image cut and fusion, cloud detection, target detection, segmentation, image compression Output: L0-L4 level remote sensing image
Onboard AI	Type: Large vision language model Parameters: 0.7 B Memory usage: < 5 GB Generation speed: 23.45 token/s Capabilities: Unified architecture for image-level, region-level and pixel-level reasoning, support 14 complex remote sensing tasks (Fig. 7)

remote sensing, named FCD-78M. Specifically, the project aggregated 90 annotated, task-specific RGB image datasets, such as Million-AID [72], RSICD [73], and DOTA [74], covering nearly all publicly available remote sensing datasets acquired from satellites, airplanes, and drones. Irrelevant datasets were filtered out after manual screening, resulting in a curated collection of 67 high-quality datasets. Data preprocessing techniques are applied to extract useful features from raw data into a standardized format, facilitating more efficient processing and analysis. While the original datasets only support a limited number of remote sensing tasks, data repurposing techniques are employed to generate additional annotations, thereby expanding the dataset's applicability across a broader range of tasks. The final dataset comprises approximately 5.6 million remote sensing images with multispatial resolution and multiview perspectives, along with 78.2 million

TABLE III
FAULT TOLERANCE MECHANISMS ACROSS SYSTEM COMPONENTS

Components	Fault tolerance mechanisms
Spaceborne computer	★ Hardware redundancy at board and components ★ ECC-protected memory ★ Radiation-hardened supervisory MCU ★ Secure boot and firmware protection ★ Latch-up detection and current limiting ★ Watchdog timers and autonomous restart
Spaceborne router	★ Cold-standby routing redundancy ★ Link redundancy with fast switchover ★ Dynamic routing during link or node failures
Spaceborne laser terminal	★ Automatic beam re-acquisition and re-tracking ★ Fallback to RF link ★ Radiation-hardened detectors and ECC
Space-based operating system	★ Container-based process isolation and health monitoring daemons ★ Application migration from faulty to healthy sate. ★ Offline autonomy when disconnected from ground ★ Data replication and state synchronization across satellites and ground

TABLE IV
COMPARISONS OF FCD-78M WITH OTHER REMOTE SENSING DATASETS

Dataset	Rep.VLM	#Images	#Annota.	#Tasks
RS5M	GeoRSLIP [75]	-	5M	1
LHRS-Align	LHRS-Bot [63]	1.15M	1.15M	1
RSICap	RSGPT [60]	2191	2191	1
MMRS-1M	EarthGPT [76]	-	1M	5
SkyEye-968k	SkyEyeGPT [77]	-	968k	4
RSVP-3M	EarthMarker [78]	-	3M	5
FCD-78M	Falcon	5.6M	78.2M	14

high-quality visual annotations, featuring hierarchical annotations with manual sampling verification to ensure quality. FCD-78M fully supports all 14 complex remote sensing tasks illustrated in Fig. 7, making it a powerful resource for developing and evaluating advanced AI models in the remote sensing domain. The comparison of FCD-78M with other major remote sensing datasets is shown in Table IV.

C. Test and Validation

Extensive testing has been carried out to validate the functionality, interoperability, and fault-tolerant performance of both the individual payloads and the 3-body computing constellation as a whole. The testbed incorporates hardware-in-the-loop (HIL) with a digital twin that accurately reproduces in-orbit conditions, enabling repeatable and controlled evaluation of system behavior under realistic space deployment scenarios before launch.

1) *Testbed*: A ground-based testbed consisting of 12 interconnected satellites, each equipped with a spaceborne intelligent computer, spaceborne router, and laser communication terminal, has been constructed to emulate in-orbit operations. The overall system architecture of the testbed is shown in Fig. 8. A high-fidelity satellite attitude and orbit control module reproduces orbital trajectories and attitude dynamics using constellation configuration parameters and real-time telemetry, providing precise positional data for constellation-level simulation and validation. Satellite-to-ground and ISLs are emulated using a channel emulator via wired connections. The former reproduces communication

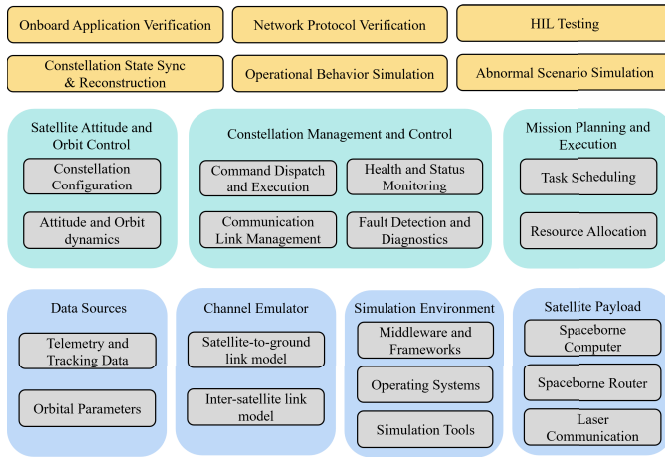


Fig. 8. System diagram of the integration test for the 12-satellite computing constellation.

accessibility between LEO satellites and ground stations by combining orbital parameters, satellite attitude data, and ground antenna characteristics to compute throughput and access windows. The latter models packet loss rates and transmission delays between satellites based on orbital configurations, attitude parameters, and laser communication terminal characteristics. A ground-based testing server, functioning as the satellite management and control center, generates telemetry, dispatches telecommands, and monitors the constellation's operational status and resource utilization. The mission planning and execution module coordinates task scheduling, orbital adjustments, and multisatellite resource allocation. In addition, a ground-based simulation and verification module enables functional and resilience testing of onboard applications and AI algorithms.

2) *Test Plans*: The test plan for the 12-satellite computing constellation encompasses both individual satellite-level and constellation-level verification across hardware, software, networking, and computing. At the hardware level, integration tests verify the compatibility and stability of the spaceborne intelligent computer, router, and laser communication unit. Routing performance has been thoroughly evaluated with respect to routing convergence, link-failure recovery, and the impact of varying delays, bit error rate (BER), and random ISL interruptions on routing stability. In-orbit computing tests benchmark throughput and efficiency under simulated orbital conditions. Single-satellite computing focuses on evaluating onboard processing of an individual satellite, while distributed computing assesses multisatellite cooperative processing. Software upload and update tests verify reliable deployment, while fault detection and recovery tests examine system resilience, including router hot-standby switching and fault isolation. Mission-level collaborative operations are also validated. Power and thermal designs are provided by the manufacturer and verified through environmental testing before launch. An overview of the test plan is provided in Table V.

The test plans were executed across a broad range of nominal and anomaly scenarios to emulate both realistic and extreme operational conditions. These scenarios span

TABLE V
OVERVIEW OF THE TEST PLAN FOR THE 12-SATELLITE
COMPUTING CONSTELLATION

Category	Description	Expected Outcome
Hardware Integration	Integration of spaceborne computer, router, and laser communication unit on each satellite	Stable operation and correct configuration of all payload components
Networking and Routing Validation	Routing convergence, link failure recovery, delay, BER, and throughput under nominal and degraded conditions; evaluate the impact of delay, BER, and ISL interruptions on routing stability	Robust network connectivity across constellation with verified routing convergence and consistency
In-Orbit Computing Performance	Benchmark computing performance of high-throughput engines and AI inference workloads	Measured computing efficiency meets or exceeds design specifications
Single-Satellite Computing	Perform functional and performance testing on a single satellite, including AI inference, data preprocessing	Verified stability reliability, and accuracy of onboard computing at the single-satellite level
Multi-Satellite Distributed Computing	Execute distributed tasks such as remote-sensing image analysis and AI inference across satellites	Coordinated execution with verified synchronization and workload distribution
Software Upload and Update	Perform remote software injection, version control, and update of onboard applications	Successful software deployment and rollback without communication interruption
Fault Detection and Recovery	Assess system fault-tolerance and redundancy mechanisms from simulated hardware software failures	Rapid fault recovery with minimal service disruption
Mission-Level Validation	Conduct single-satellite and multi-satellite collaborative tasks such as remote sensing and scientific data processing	Functional and performance compliance with mission design requirements

typical mission operations such as satellite attitude and orbit control, data acquisition and transmission, and intersatellite networking, as well as stress tests under degraded conditions, including link interruptions, hardware faults, and software anomalies. In total, 247 test cases were executed using a combined HIL and digital-twin simulation environment, all of which passed successfully, demonstrating the constellation's technical feasibility and readiness for operational deployment in space.

3) *Ground Validation and In-Orbit Experiments*: The performance of the trained Falcon model has been evaluated and compared against both general-purpose vision-language models (such as MiniGPTv2 [79], MiniCPM [80], QwenChat [81], and Florence-2 [71]) and specialized remote sensing VLMs (such as RemoteCLIP [67], GeoChat [68], and LHRs-Bot [63]) across 14 remote sensing tasks, as illustrated in Fig. 7. Falcon demonstrated impressive performance with only 0.7 billion parameters across more than 100 test datasets, exhibiting strong compositional perception and reasoning capabilities. Fig. 9 presents an example of Falcon's output on

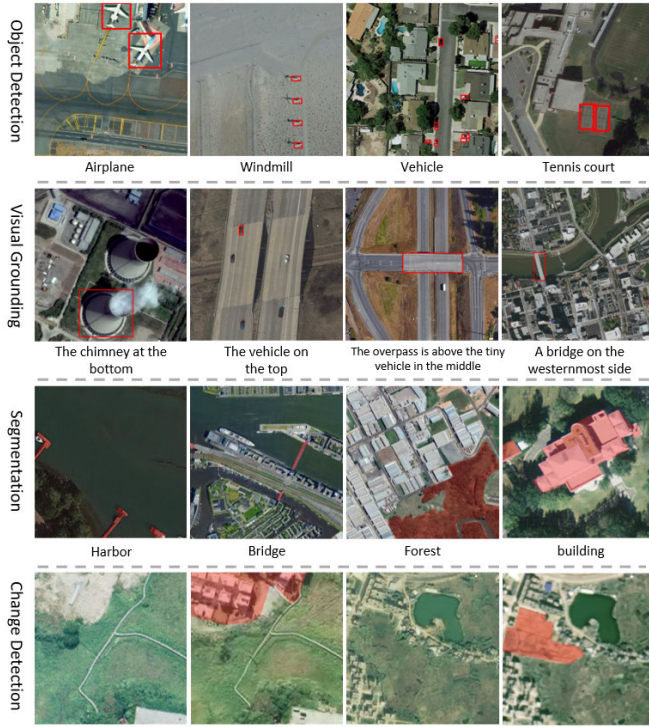


Fig. 9. Visualization of Falcon's output on tasks of object detection, visual grounding, segmentation, and change detection.

pixel-level tasks such as segmentation and change detection, where the model accurately segmented complex targets and successfully identified changes between two similar images. A detailed and comprehensive evaluation of Falcon's performance is provided in [69].

In-orbit object detection experiments have also been performed in the initial 3-body computing constellation with both single-satellite computing and multisatellite distributed computing. The workflow is structured as follows: Users begin by submitting natural language instructions, which are parsed, extracted, and preprocessed by Tianhui, which is a 7B LLM deployed on the ground. This model interacts with the space-based distributed system for task scheduling and automatically generates the corresponding instruction files that specify the precise actions to be executed by the participating satellites, before uploading them to a satellite connected to a ground station. Upon receiving the instruction file, the satellite parses its contents and inputs them into the onboard model, which generates a task plan tailored to the specified requirements. If the task involves coordination with other satellites, the instruction file is transmitted via ISLs to the relevant satellite, where a localized task plan is generated. Guided by the task plan, the onboard model sequentially invokes the appropriate tools from the available onboard toolset and dynamically adjusts the execution process based on real-time feedback, considering factors such as computational load, network status, energy availability, sensor type, and satellite positioning. Once the task is completed, the resulting data is packaged and transmitted back to the ground via the space-based distributed file system. If the computation was performed on a remote

TABLE VI
IN-ORBIT MULTISATELLITE REMOTE SENSING TASK TIMELINE

Step	Actions	Duration
1	Ground LLM parses instruction and generates instruction files	18 s
2	Upload instruction files from GS to Sat-1 (ground-sat link)	62 s
3	Sat-1 parses files and compiles local/relay plan	17 s
4	Relay instruction from Sat-1 to Sat-3 via ISL (APT closed, small file + control)	11 s
5	Sat-3 parses files & compiles local plan	14 s
6	Orbital alignment wait until target is in view	82 s
7	Sat-3 slews to target and stabilizes	87 s
8	Image acquisition	12 s
9	On-orbit preprocessing (radiometric/geo correction, tiling)	67 s
10	On-board AI inference (object detection and post-processing)	98 s
11	Package results and metadata (~ 500 MB)	19 s
12	Results relay from Sat-3 to Sat-1 via ISL	21 s
13	Downlink results from Sat-1 to GS (window open, ~ 500 MB@150–300 Mbps incl. overhead)	45 s
14	Ground analysis, catalog, visualization	18 s
Total Response Time:		569 s

satellite, the results are first relayed via ISLs to a satellite with direct ground station connectivity before being downloaded. Upon receipt, the ground system parses and visualizes the data through a user interface. Users can interact with the received remote sensing images via a visual question-and-answer interface, where they can perform intelligent functionalities such as scene classification, target localization, object detection and counting, image interpretation, regional queries, and other advanced forms of semantic interaction tailored to the user's needs.

Specifically, an experiment of in-orbit multisatellite remote-sensing task for object detection is illustrated in Table VI, with detailed step-by-step timings. A user instruction is first parsed at the ground, and an execution plan is generated that assigns Satellite 3 to perform image acquisition and onboard AI inference to minimize latency. The instruction files are uploaded to Satellite 1 and then forwarded to Satellite 3 via ISLs for execution. The processed results are relayed back to Satellite 1, whose downlink window remains open, and subsequently transmitted to the ground for visualization and analysis. The end-to-end response time is approximately 9 min, which is substantially shorter than the hour-level completion time of a comparable single-satellite workflow (approximately 90% reduction), where the revisit time of an individual satellite in the initial 3-body constellation is roughly 90 min with only one available ground station.

The end-to-end response time is a critical key performance indicator (KPI) for remote sensing missions, particularly in time-sensitive scenarios such as disaster monitoring and maritime surveillance. Conventional remote sensing workflows are primarily limited by two dominant latency bottlenecks: satellite revisit time and downlink latency. The former is determined by the number of orbital periods the satellite has to travel before it can complete the required actions, while the latter scales with raw data volume and is further constrained by intermittent ground-station visibility. Multisatellite distributed

computing alleviates these constraints by performing AI inference directly in orbit, such that only processed results, often less than 1% of the raw data volume, need to be transmitted to the ground, reducing communication latency by an order of magnitude. In addition, cooperative execution across multiple satellites further shortens response time: upon receiving a task, it can be forwarded via ISLs to the satellite that will image the target earliest and perform onboard processing. As a result, distributed execution simultaneously minimizes orbital latency and downlink delay, enabling rapid access to actionable results that are unattainable with traditional single-satellite or ground-centric architectures. This approach is particularly advantageous in scenarios where ground stations are sparse or geographically limited, in which waiting times for downlink opportunities dominate overall response time.

VI. RESEARCH OPPORTUNITIES

As space computing is still in its early stages, the implementation challenges identified in Section IV highlight the need for further research. Building on these insights, this section outlines promising research directions that address current limitations and will help shape the future development of space computing.

A. AI-Hardware Integration and Training

Although current implementations of spaceborne computer and space-based vision-language foundation models, such as Falcon, can effectively balance computational performance with satellite resource constraints, tradeoffs remain in terms of representational richness and adaptability across diverse tasks. Future work should investigate the tight integration of AI models with the underlying hardware platform, where reconfigurable platforms like FPGAs offer high performance, energy efficiency, radiation resilience, and inherent flexibility that can be tailored to the unique requirements of spaceborne AI tasks. Nonetheless, integrating AI models with FPGAs remains challenging owing to resource constraints, algorithm-hardware mismatches, parallelization inefficiencies, and the lack of automated design tools [82].

Addressing these issues requires joint progress at both the algorithmic and hardware levels. Algorithm-level optimizations, such as sparsification, model restructuring, low-rank approximation, and quantization, can reduce computational complexity, while hardware-level advances, including optimized dataflow and parallel computing architectures, further enhance efficiency [83]. Algorithm-hardware co-design is therefore essential, aligning model architectures with hardware constraints and tailoring hardware to accelerate specific workloads. Standardized interfaces between AI frameworks and onboard hardware are also critical for portability, enabling models trained on the ground to be seamlessly deployed in orbit. In addition, radiation-aware AI model design, which incorporates robustness against bit flips and error-tolerant mechanisms, is vital for ensuring reliability.

The performance of large AI models, such as Falcon, heavily relies on the availability of extensive and diverse datasets, as well as specialized training techniques tailored to

the unique challenges of the space domain. However, integrating heterogeneous data from diverse network components is nontrivial, involving issues such as data format compatibility, scalability, and data privacy and security. Addressing these challenges requires adaptive data fusion techniques capable of dynamically integrating and analyzing data from disparate sources. To alleviate annotation bottlenecks, cross-modal, multitask, fine-grained labeling engines that integrate expert models with multiexpert decision optimization should be developed, thereby generating large-scale annotated datasets at significantly reduced cost. At the same time, mitigating the computational burden of large AI models will require continued exploration of techniques such as model compression, pruning, knowledge distillation, and distributed inference across multiple satellites. Although LoRA-based incremental updates help mitigate communication overhead, real-time synchronization across satellites remains challenging under intermittent links. Collectively, these advances will foster tighter integration between AI algorithms and computing substrates, enabling more scalable, resilient, and adaptive AI ecosystems operating across entire constellations.

B. Networking

Given the current limitations of the implemented spaceborne router, particularly its restricted scalability in large-scale constellations and lack of full support for integrated satellite-terrestrial networking, future developments will need to incorporate more advanced networking paradigms such as segment routing over IPv6 (SRv6) for seamless satellite-terrestrial integration and AI-driven adaptive routing. SRv6, which combines the scalability of IPv6 addressing with the flexibility of source routing, offers a promising approach to achieving seamless end-to-end integration between satellite and terrestrial networks. Its deployment in LEO constellations, however, requires lightweight protocol designs that minimize computational and signaling overhead while ensuring efficient route calculation, seamless handover across unstable ISLs, and rapid recovery from single-node failures. Another promising direction is the integration of AI for predictive modeling and adaptive routing [84]. This is of particular interest with the increasing complexity of future satellite constellations and the trend that satellites are equipped with more powerful computational capabilities. Traditional routing algorithms are less efficient and often suffer from high overhead in recalculating routes caused by frequent topology changes. By leveraging historical data and monitoring real-time parameters such as link quality, resource utilization, and traffic patterns, AI can proactively predict link stability, detect anomalies, optimize traffic flow, and dynamically adjust routing strategies accordingly. This enables autonomous self-healing capabilities that not only reduce reliance on ground control but also enhance resilience and efficiency.

Furthermore, to address the power and cost inefficiencies of deploying routing modules and laser communication terminals as separate payloads, future research should investigate integrated router-laser communication architectures that tightly couple networking and physical-layer functions. On the hardware level, the use of high-performance dedicated

network switching chips combined with embedded CPUs will ensure high-speed packet forwarding and efficient protocol processing, while the software layer can employ open-source network operating systems with abstracted software interfaces to enable flexible definition of network functions, optimized traffic scheduling, and load balancing under high-throughput conditions. Reliability enhancements can be achieved through automated fault detection, containerized system design, and real-time network monitoring, ensuring continuous service availability. Furthermore, triple-redundancy protection mechanisms, including primary/backup cold standby switching, periodic data verification, ECC memory error correction, and over-the-air (OTA) firmware upgrades, should be incorporated to ensure robust data exchange and system resilience. Together, these advancements will enable an intelligent, energy-efficient, and highly reliable spaceborne optical networking infrastructure that seamlessly integrates routing and laser communication functionalities.

C. Embodied Intelligent Satellite

While Falcon demonstrates the feasibility of lightweight vision–language foundation models for remote sensing, future research should move beyond task-specific applications toward the realization of embodied intelligent satellites capable of autonomous perception, reasoning, and action. With advancements in neuromorphic processors and AI acceleration hardware, it is becoming increasingly feasible to deploy sophisticated AI models directly onboard satellites, enabling them to perform complex tasks autonomously with enhanced adaptability and situational awareness.

Embodied intelligent satellites represent a paradigm shift toward fully autonomous and intelligent space systems that are not only reactive but also proactive and adaptive [85]. Unlike traditional satellites that rely heavily on ground-based control and preprogrammed instructions, embodied intelligent satellites leverage AI to prioritize constellation-wide autonomy, which includes understanding mission objectives, perceiving their environment, planning cooperatively across multiple satellites, and executing tasks with minimal human intervention. In addition, such satellites should be capable of intelligent energy and thermal management, predictive fault detection, and self-healing. Collectively, these advancements establish the principle of “embodied intelligence,” transforming satellites from passive data collectors into active agents. This concept is inspired by the idea that intelligence emerges through continuous interaction between a system’s physical body and its environment.

AI is the key enabler of this transformation, providing the cognitive capabilities required for autonomous operation. Its integration into space infrastructure can significantly enhance mission resilience, reduce operational latency, and support more ambitious undertakings such as deep space exploration and fully autonomous satellite constellations.

D. Testbeds and Digital Twins

Conducting experiments directly on physical infrastructure can be both challenging and costly, particularly for large-scale satellite constellations. To address the scalability and

cost constraints of physical payloads, an emerging area of interest is the development of hybrid large-scale constellation emulation platforms that integrate both physical and virtual payloads. The system shall investigate the fusion of virtual and physical satellites, enabling efficient validation of routing algorithms and multisatellite collaboration schemes across constellations comprising hundreds of nodes. Virtual satellites can be instantiated with different levels of isolation, from VMs and containers to micro-VMs and network namespaces, allowing for flexible and scalable deployment according to experiment requirements. This hybrid architecture provides a cost-effective and technically scalable solution for large-scale constellation emulation.

Digital twins, on the other hand, facilitate real-time monitoring, simulation, predictive analytics, and scenario testing without the limitations of physical deployments by dynamically mirroring the state and behavior of their real-world counterparts and operational environments. A high-fidelity constellation model, driven by both knowledge and data, is crucial for accurately simulating and optimizing the performance of satellite networks. To achieve this, the model must integrate multiple components to enhance the realism and precision of satellite behavior. Specifically, comprehensive space-to-ground and intersatellite communication models must be developed, accounting for environmental factors such as rain attenuation, atmospheric absorption, solar interference, and other harsh space conditions to improve link simulation accuracy. The simulation of radiation effects on hardware presents a significant challenge, yet it is a critical consideration for space-based applications. Additionally, an energy generation and management model should be established, incorporating solar panel power generation, battery charge/discharge characteristics, and satellite payload power consumption profiles, to analyze energy balance and thermal management. This integrated approach allows for the precise reconstruction and emulation of in-orbit satellite behaviors, including payload operations, satellite operational states, and constellation networking dynamics.

Developing embodied AI demands extensive training data to ensure safe and effective interaction with real environments, making cost-efficient simulated environments essential. However, transferring models, algorithms, or control strategies trained in a simulated environment to real systems remains challenging [86], as simplified models fail to capture the complexity of real-world conditions, especially in large-scale satellite constellations. Integrating digital twins with embodied AI offers a transformative solution. Digital twins create high-fidelity virtual representations, while Embodied AI leverages them for perception, learning, and interaction. This synergy allows embodied agents to train in digital replicas of real-world environments, thereby reducing training costs, improving safety, and enhancing adaptability in real-world operations.

VII. CONCLUSION

Space computing represents a transformative paradigm shift, enabling the onboard processing of space-native data,

autonomous decision-making, and mission adaptability without reliance on ground stations. This capability is essential not only for alleviating data transmission bottlenecks between satellites and ground infrastructure but also for supporting advanced space missions such as Earth observation and deep space exploration. In this article, we first introduce the concept of space computing and outline its system architecture and key enabling technologies, including spaceborne computers, intersatellite laser communications, spaceborne routers, distributed operating systems, and onboard AI models. We then present the design and implementation of the 3-body computing constellation, along with the in-orbit experiments on multisatellite remote sensing tasks, demonstrating that its end-to-end response time is substantially shorter than the hour-level latency associated with comparable single-satellite workflows. Finally, we discuss several promising research directions aimed at advancing intelligent, scalable, and resilient space computing infrastructure, covering AI–hardware co-design, networking, digital-twin-based testbeds, and the envisioned embodied intelligent satellite. As space computing is still in its infancy, this article highlights some critical challenges that we believe will shape the next wave of innovation and unlock the full potential of space computing in the years ahead.

REFERENCES

- [1] Z. Zhang, W. Zhang, and F.-H. Tseng, "Satellite mobile edge computing: Improving QoS of high-speed satellite-terrestrial networks using edge computing techniques," *IEEE Netw.*, vol. 33, no. 1, pp. 70–76, Jan. 2019.
- [2] NASA. (2024). *State-of-The-Art of Small Spacecraft Technology*. [Online]. Available: <https://www.nasa.gov/smallsat-institute/sst-soa/>
- [3] D. Bhattacharjee, S. Kassing, M. Licciardello, and A. Singla, "In-orbit computing: An outlandish thought experiment?," in *Proc. 19th ACM Workshop Hot Topics Netw.*, 2020, pp. 197–204.
- [4] Y. Li, M. Wang, K. Hwang, Z. Li, and T. Ji, "LEO satellite constellation for global-scale remote sensing with on-orbit cloud AI computing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 9369–9381, 2023.
- [5] B. Denby, K. Chintalapudi, R. Chandra, B. Lucia, and S. Noghabi, "Kodan: Addressing the computational bottleneck in space," in *Proc. 28th ACM Int. Conf. Architectural Support Program. Lang. Operating Syst.*, Mar. 2023, pp. 392–403.
- [6] S. Milan, "KuaFu: Exploring the Sun–Earth connection," *Astron. Geophys.*, vol. 53, no. 4, pp. 421–424, Aug. 2012.
- [7] C. W. Royer, "Fog computing in optically networked space constellations," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 40, no. 1, pp. 22–33, Jan. 2025.
- [8] Y. Zengshan et al., "A comprehensive survey of orbital edge computing: Systems, applications, and algorithms," *Chin. J. Aeronaut.*, vol. 38, no. 7, Jul. 2025, Art. no. 103316.
- [9] X. Ma, M. Xu, Q. Li, Y. Li, A. Zhou, and S. Wang, *5G Edge Computing: Technologies, Applications and Future Visions*. Cham, Switzerland: Springer, 2024.
- [10] S. Wang, Q. Li, M. Xu, X. Ma, A. Zhou, and Q. Sun, "Tiansuan constellation: An open research platform," in *Proc. IEEE Int. Conf. Edge Comput. (EDGE)*, Sep. 2021, pp. 94–101.
- [11] C. Wang, Y. Zhang, Q. Li, A. Zhou, and S. Wang, "Satellite computing: A case study of cloud-native satellites," in *Proc. IEEE Int. Conf. Edge Comput. Commun. (EDGE)*, Jul. 2023, pp. 262–270.
- [12] Y. Lin et al., "Satellite-MEC integration for 6G Internet of Things: Minimal structures, advances, and prospects," *IEEE Open J. Commun. Soc.*, vol. 5, pp. 3886–3903, 2024.
- [13] Q.-V. Pham et al., "Aerial computing: A new computing paradigm, applications, and challenges," *IEEE Internet Things J.*, vol. 9, no. 11, pp. 8339–8363, Jun. 2022.
- [14] Z. Shen et al., "A survey of next-generation computing technologies in space-air-ground integrated networks," *ACM Comput. Surv.*, vol. 56, no. 1, pp. 1–40, Aug. 2023.
- [15] S. Wang and Q. Li, "Satellite computing: Vision and challenges," *IEEE Internet Things J.*, vol. 10, no. 24, pp. 22514–22529, Dec. 2023.
- [16] D. Bhattacharjee and A. Singla, "Network topology design at 27,000 km/hour," in *Proc. 15th Int. Conf. Emerg. Netw. Exp. Technol.*, Dec. 2019, pp. 341–354.
- [17] D. Bhattacharjee et al., "Gearing up for the 21st century space race," in *Proc. ACM Workshop Hot Topics Netw.*, 2018, pp. 113–119.
- [18] S. Kassing, D. Bhattacharjee, A. B. Águas, J. E. Saethre, and A. Singla, "Exploring the 'internet from space' with Hypatia," in *Proc. ACM Internet Meas. Conf.*, Oct. 2020, pp. 214–229.
- [19] G. Giuliani, T. Klenze, M. Legner, D. Basin, A. Perrig, and A. Singla, "Internet backbones in space," *ACM SIGCOMM Comput. Commun. Rev.*, vol. 50, no. 1, pp. 25–37, Mar. 2020.
- [20] Y. Hauri, D. Bhattacharjee, M. Grossmann, and A. Singla, "Internet from space' without inter-satellite links," in *Proc. 19th ACM Workshop Hot Topics Netw.*, Nov. 2020, pp. 205–211.
- [21] B. Denby and B. Lucia, "Orbital edge computing: Nanosatellite constellations as a new class of computer system," in *Proc. 25th Int. Conf. Architectural Support Program. Lang. Operating Syst.*, Mar. 2020, pp. 939–954.
- [22] Q. Li et al., "Service coverage for satellite edge computing," *IEEE Internet Things J.*, vol. 9, no. 1, pp. 695–705, Jan. 2022.
- [23] M. Y. Abdelsadek et al., "Future space networks: Toward the next giant leap for humankind," *IEEE Trans. Commun.*, vol. 71, no. 2, pp. 949–1007, Feb. 2023.
- [24] A. Cratere, L. Gagliardi, G. A. Sanca, F. Golmar, and F. Dell'Olio, "On-board computer for CubeSats: State-of-the-art and future trends," *IEEE Access*, vol. 12, pp. 99537–99569, 2024.
- [25] M. Qasaimeh et al., "Benchmarking vision kernels and neural network inference accelerators on embedded platforms," *J. Syst. Archit.*, vol. 113, Feb. 2021, Art. no. 101896.
- [26] Google-Cloud. *Accelerate AI Development With Google Cloud Tpus*. Accessed: May 2025. [Online]. Available: <https://cloud.google.com/tpu>
- [27] Microsoft-Brainwave. *Project Brainwave*. Accessed: May 2025. [Online]. Available: <https://www.microsoft.com/en-us/research/project/project-brainwave/>
- [28] AWS-Inferentia. *AWS Inferentia—AI Chip*. Accessed: Apr. 2025. [Online]. Available: <https://aws.amazon.com/cn/ai/machine-learning/inferentia/>
- [29] F. Akopyan et al., "TrueNorth: Design and tool flow of a 65 mW 1 million neuron programmable neurosynaptic chip," *IEEE Trans. Comput.-Aided Design Integr. Circuits Syst.*, vol. 34, no. 10, pp. 1537–1557, Oct. 2015.
- [30] AMD-Xilinx. (2024). *AMD ZYNQ 7000 SOC's*. Accessed: Feb. 2025. [Online]. Available: <https://www.amd.com/en/products/adaptive-socs-and-fpgas/soc/zynq-7000.html>
- [31] Microchip-Technology. (2024). *Smartfusion 2 SOC FPGAS*. [Online]. Available: <https://www.microchip.com/enus/products/fpgas-and-plds/system-on-chip-fpgas/smartfusion-2-fpgas>
- [32] Z. E. Khatib et al., "LoRa-enabled GPU-based CubeSat YOLO object detection with hyperparameter optimization," in *Proc. Int. Symp. Netw., Comput. Commun. (ISNCC)*, Jul. 2022, pp. 1–4.
- [33] B. Todd and S. Uznański, "Radiation risks and mitigation in electronic systems," in *Proc. CAS- CERN Accel. School, Power Converters*, Jun. 2015, pp. 245–263.
- [34] J. Prinzie, F. M. Simanjuntak, P. Leroux, and T. Prodromakis, "Low-power electronic technologies for harsh radiation environments," *Nature Electron.*, vol. 4, no. 4, pp. 243–253, Apr. 2021.
- [35] A. J. Tylka, W. F. Dietrich, P. R. Boberg, E. C. Smith, and J. H. Adams, "Single event upsets caused by solar energetic heavy ions," *IEEE Trans. Nucl. Sci.*, vol. 43, no. 6, pp. 2758–2766, Dec. 1996.
- [36] P. J. Botma, A. Barnard, and W. H. Steyn, "Low cost fault tolerant techniques for nano/pico-satellite applications," in *Proc. Africon*, Sep. 2013, pp. 1–5.
- [37] C. M. Fuchs, N. M. Murillo, A. Plaat, E. van der Kouwe, D. Harsono, and T. P. Stefanov, "Fault-tolerant nanosatellite computing on a budget," in *Proc. 18th Eur. Conf. Radiat. Its Effects Compon. Syst. (RADECS)*, Sep. 2018, pp. 1–8.
- [38] D. W. Caldwell and D. A. Rennels, "A minimalist fault-tolerant micro-controller design for embedded spacecraft computing," *J. Supercomput.*, vol. 16, nos. 1–2, pp. 7–25, May 2000.
- [39] M. Yang, G. Hua, Y. Feng, and J. Gong, "Chapter 5: Software fault-tolerance techniques," in *Fault-Tolerance Techniques for Spacecraft Control Computers*. Hoboken, NJ, USA: Wiley, 2017, ch. 5, pp. 151–178.
- [40] I. Sünter et al., "Firmware updating systems for nanosatellites," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 31, no. 5, pp. 36–44, May 2016.

- [41] P. Dobiáš, E. Casseau, and O. Sinnen, "Improving the CubeSat reliability thanks to a multiprocessor system using fault tolerant online scheduling," *Microprocessors Microsystems*, vol. 85, Sep. 2021, Art. no. 104312.
- [42] N. Murphy and M. Barr, "Watchdog timers," *Embedded Syst. Program.*, vol. 14, pp. 79–80, Oct. 2001.
- [43] A. V. Dias, J. A. Pomilio, and S. Finco, "A current limiting switch for applications in space power systems," in *Proc. IEEE Southern Power Electron. Conf. (SPEC)*, Dec. 2017, pp. 1–6.
- [44] A. Gupta and S. Nagpal, "Design and evaluation of 10-Gbps inter-satellite optical wireless communication link for improved performance," *J. Opt. Commun.*, vol. 38, no. 2, pp. 195–199, May 2017.
- [45] M. Toyoshima, "Recent trends in space laser communications for small satellites and constellations," *J. Lightw. Technol.*, vol. 39, no. 3, pp. 693–699, Feb. 1, 2021.
- [46] A. Panahi and A. A. Kazemi, "High speed laser communication network for satellite systems," *Proc. SPIE*, vol. 8026, May 2011, Art. no. 80260L.
- [47] A. A. Kazemi, "Intersatellite laser communication systems for harsh environment of space," *Proc. SPIE*, vol. 8720, May 2013, Art. no. 872010.
- [48] A. Bagrov and V. Lukin, "Laser optical communication systems with space transmitters," *Proc. SPIE*, vol. 6160, pp. 221–227, Apr. 2006.
- [49] E. Ekici, I. F. Akyildiz, and M. D. Bender, "A distributed routing algorithm for datagram traffic in LEO satellite networks," *IEEE/ACM Trans. Netw.*, vol. 9, no. 2, pp. 137–147, Apr. 2001.
- [50] K. Li, C. Jiang, Y. Wang, D. Zhao, Q. Li, and Y. Jiang, "Achieving scalable and efficient routing in LEO constellations," in *Proc. IEEE Int. Conf. Satell. Comput. (Satellite)*, Nov. 2023, pp. 25–30.
- [51] T. Pan, T. Huang, X. Li, Y. Chen, W. Xue, and Y. Liu, "OPSPF: Orbit prediction shortest path first routing for resilient LEO satellite networks," in *Proc. IEEE Int. Conf. Commun. (ICC)*, May 2019, pp. 1–6.
- [52] X. Zhang, Y. Yang, M. Xu, and J. Luo, "ASER: Scalable distributed routing protocol for LEO satellite networks," in *Proc. IEEE 46th Conf. Local Comput. Netw. (LCN)*, Oct. 2021, pp. 65–72.
- [53] J. Eickhoff, *Onboard Computers, Onboard Software and Satellite Operations: An Introduction*. Cham, Switzerland: Springer, 2012.
- [54] M. Centenaro, C. E. Costa, F. Granelli, C. Sacchi, and L. Vangelista, "A survey on technologies, standards and open challenges in satellite IoT," *IEEE Commun. Surv. Tut.*, vol. 23, no. 3, pp. 1693–1720, 3rd Quart., 2021.
- [55] M. Fayyaz, T. Vladimirova, and J.-M. Caujolle, "Adaptive middleware design for satellite fault-tolerant distributed computing," in *Proc. NASA/ESA Conf. Adapt. Hardw. Syst. (AHS)*, Jun. 2012, pp. 23–30.
- [56] V. Kothari, E. Liberis, and N. D. Lane, "The final frontier: Deep learning in space," in *Proc. 21st Int. Workshop Mobile Comput. Syst. Appl.*, Mar. 2020, pp. 45–49.
- [57] Y. Xiong, Y. Sun, L. Xing, and Y. Huang, "Extend cloud to edge with KubeEdge," in *Proc. IEEE/ACM Symp. Edge Comput. (SEC)*, Oct. 2018, pp. 373–377.
- [58] T. Pfandzelter, J. Hasenburg, and D. Bernbach, "Towards a computing platform for the LEO edge," in *Proc. 4th Int. Workshop Edge Syst., Anal. Netw.*, Apr. 2021, pp. 43–48.
- [59] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 296–307, Jan. 2020.
- [60] Y. Hu, J. Yuan, C. Wen, X. Lu, and X. Li, "RSGPT: A remote sensing vision language model and benchmark," 2023, *arXiv:2307.15266*.
- [61] X. Sun et al., "FAIR1M: A benchmark dataset for fine-grained object recognition in high-resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 184, pp. 116–130, Feb. 2022.
- [62] Z. Yuan et al., "Exploring a fine-grained multiscale method for cross-modal remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, May 2022, Art. no. 4404119.
- [63] D. Muhtar, Z. Li, F. Gu, X. Zhang, and P. Xiao, "LHRS-bot: Empowering remote sensing with VGI-enhanced large multimodal language model," in *Proc. 18th Eur. Conf. Comput. Vis. (ECCV)*, Nov. 2024, pp. 440–457.
- [64] Y. Zhan, Z. Xiong, and Y. Yuan, "RSVG: Exploring data and models for visual grounding on remote sensing data," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, Feb. 2023, Art. no. 5604513.
- [65] X. Li, D. Hong, and J. Chanussot, "S2MAE: A spatial-spectral pre-training foundation model for spectral remote sensing data," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 27696–27705.
- [66] X. Sun et al., "RingMo: A remote sensing foundation model with masked image modeling," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, Jul. 2022, Art. no. 5612822.
- [67] F. Liu et al., "RemoteCLIP: A vision language foundation model for remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, Apr. 2024, Art. no. 5622216.
- [68] K. Kuckreja, M. S. Danish, M. Naseer, A. Das, S. Khan, and F. S. Khan, "GeoChat: Grounded large vision-language model for remote sensing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 27831–27840.
- [69] K. Yao et al., "Falcon: A remote sensing vision-language foundation model (technical report)," 2025, *arXiv:2503.11070*.
- [70] W. Wang et al., "VisionLLM: Large language model is also an open-ended decoder for vision-centric tasks," in *Proc. 37th Int. Conf. Neural Inf. Process. Syst. (NIPS)*, Dec. 2023, pp. 61501–61513.
- [71] B. Xiao et al., "Florence-2: Advancing a unified representation for a variety of vision tasks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 4818–4829.
- [72] Y. Long et al., "On creating benchmark dataset for aerial image interpretation: Reviews, guidances, and million-AID," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4205–4230, 2021.
- [73] X. Lu, B. Wang, X. Zheng, and X. Li, "Exploring models and data for remote sensing image caption generation," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2183–2195, Apr. 2018.
- [74] G.-S. Xia et al., "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3974–3983.
- [75] Z. Zhang, T. Zhao, Y. Guo, and J. Yin, "RS5M and GeoRSClip: A large-scale vision-language dataset and a large vision-language model for remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, Sep. 2024, Art. no. 5642123.
- [76] W. Zhang, M. Cai, T. Zhang, Y. Zhuang, and X. Mao, "EarthGPT: A universal multimodal large language model for multisensor image comprehension in remote sensing domain," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, Jun. 2024, Art. no. 5917820.
- [77] Y. Zhan, Z. Xiong, and Y. Yuan, "SkyEyeGPT: Unifying remote sensing vision-language tasks via instruction tuning with large language model," *ISPRS J. Photogramm. Remote Sens.*, vol. 221, pp. 64–77, Mar. 2025.
- [78] W. Zhang, M. Cai, T. Zhang, Y. Zhuang, J. Li, and X. Mao, "EarthMarker: A visual prompting multimodal large language model for remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, Dec. 2024, Art. no. 5604219.
- [79] D. Zhu, J. Chen, X. Shen, X. Li, and M. Elhoseiny, "MiniGPT-4: Enhancing vision-language understanding with advanced large language models," 2023, *arXiv:2304.10592*.
- [80] S. Hu et al., "MiniCPM: Unveiling the potential of small language models with scalable training strategies," 2024, *arXiv:2404.06395*.
- [81] J. Bai et al., "Qwen-VL: A frontier large vision-language model with versatile abilities," 2023, *arXiv:2308.12966*.
- [82] L. Riha, J. Le Moigne, and T. El-Ghazawi, "Optimization of selected remote sensing algorithms for many-core architectures," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 12, pp. 5576–5587, Dec. 2016.
- [83] F. Liu, H. Li, W. Hu, and Y. He, "Review of neural network model acceleration techniques based on FPGA platforms," *Neurocomputing*, vol. 610, Dec. 2024, Art. no. 128511.
- [84] S. Javaid, R. A. Khalil, N. Saeed, B. He, and M.-S. Alouini, "Leveraging large language models for integrated satellite-aerial-terrestrial networks: Recent advances and future directions," *IEEE Open J. Commun. Soc.*, vol. 6, pp. 399–432, 2025.
- [85] G. Agrim, S. Silvio, G. Surya, and F. Li, "Embodied intelligence via learning and evolution," *Nature Commun.*, vol. 12, pp. 1–12, Oct. 2021.
- [86] J. Li and S. X. Yang, "Digital twins to embodied artificial intelligence: Review and perspective," *Intell. Robot.*, vol. 5, no. 1, pp. 202–227, Mar. 2025.

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